

Day 1

Electromagnetic Theory

Applications:-

→ Glucose meter, transformers, Motors, ICs, Wireless, Satellites, mobile phone, Bullet train,

Gauss's law for magnetism:-

$$\vec{\nabla} \cdot \vec{E} = \rho / \epsilon_0$$

$$\oint_S \vec{D} \cdot d\vec{s} = Q_{enc}$$

Faraday's law

$$\oint \vec{\nabla} \times \vec{E} = - \frac{d\vec{B}}{dt}$$

$$\oint_C \vec{E} \cdot d\vec{l} = - \frac{d}{dt} \oint_S \vec{B} \cdot \hat{n} da$$

Ampere's law

$$\vec{\nabla} \times \vec{B} = \mu_0 \vec{j}$$

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 \oint_S \vec{j} \cdot d\vec{A}$$

Scalar

- only magnitude
- Uni dimensional

- Scalars can be divided by another scalar

- Time, mass, length

Example

Vector

- Magnitude & direction
- More than one dimension
- Vectors cannot be divided by another vector

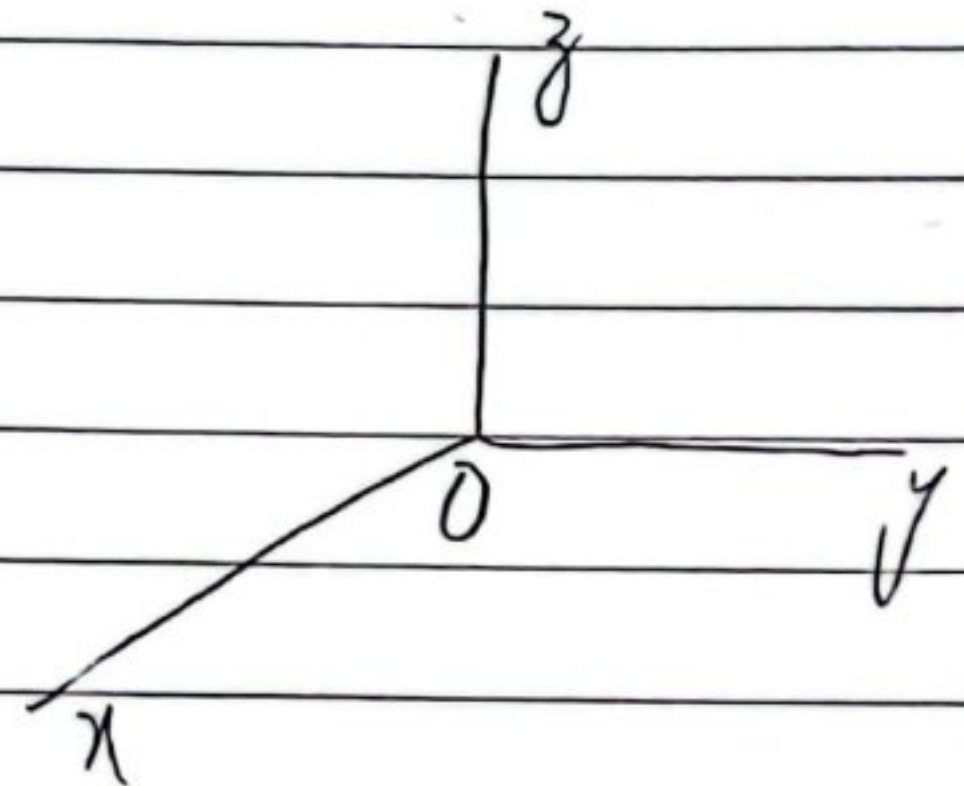
- Velocity, Weight, Force

Basis of vector:-

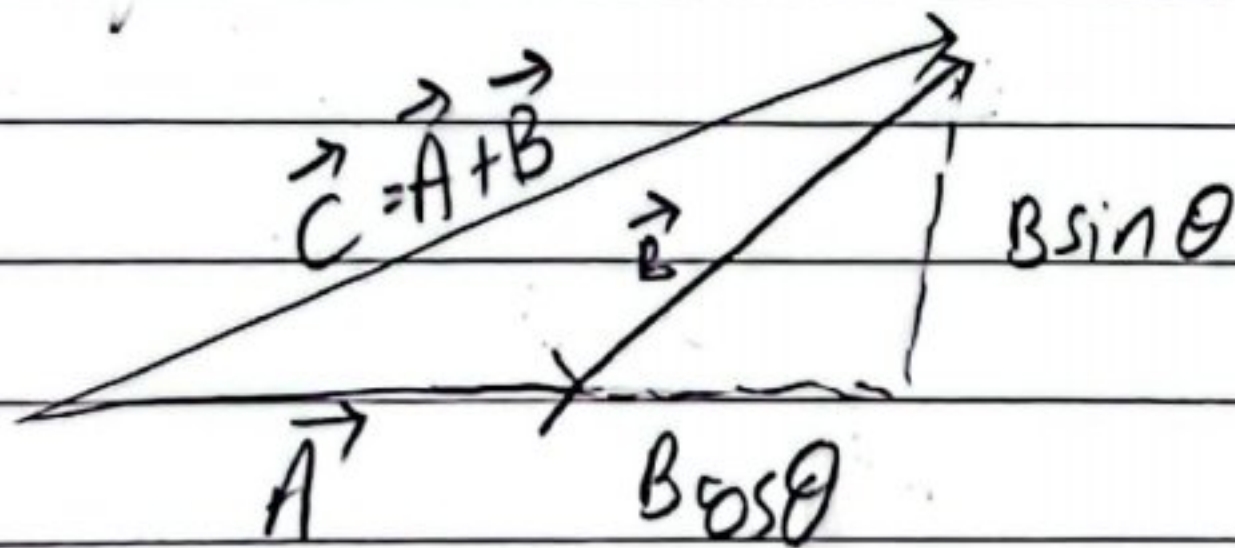
$$\vec{A} = A_x \hat{i} + A_y \hat{j} + A_z \hat{k}$$

$$|\vec{A}| = \sqrt{A_x^2 + A_y^2 + A_z^2}$$

$$\hat{A} = \frac{\vec{A}}{|\vec{A}|} = \hat{i}_A$$



∴ Vector Addition (head to tail rule)
(11grom)



$$C^2 = (A + B \cos \theta)^2 + (B \sin \theta)^2$$

$$C^2 = A^2 + B^2 \cos^2 \theta + 2AB \cos \theta + B^2 \sin^2 \theta$$

$$C^2 = A^2 + 2AB \cos \theta + B^2 (\cos^2 \theta + \sin^2 \theta)$$

$$C^2 = A^2 + B^2 + 2AB \cos \theta$$

Dot product:-

$$\vec{C} = \vec{A} \cdot \vec{B}$$

$$\vec{A} = A_x \hat{i} + A_y \hat{j} + A_z \hat{k}$$

$$\vec{B} = B_x \hat{i} + B_y \hat{j} + B_z \hat{k}$$

$$\vec{A} \cdot \vec{B} = |\vec{A}| |\vec{B}| \cos \theta$$

$$= A_x B_x + A_y B_y + A_z B_z$$

Cross Product

(To find direction use thumb rule)
∴ right hand rule

$$\vec{Y} = \vec{A} \times \vec{B} = |\vec{A}| |\vec{B}| \sin \theta$$

$$\vec{Y} = \vec{A} \times \vec{B} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ A_x & A_y & A_z \\ B_x & B_y & B_z \end{vmatrix}$$

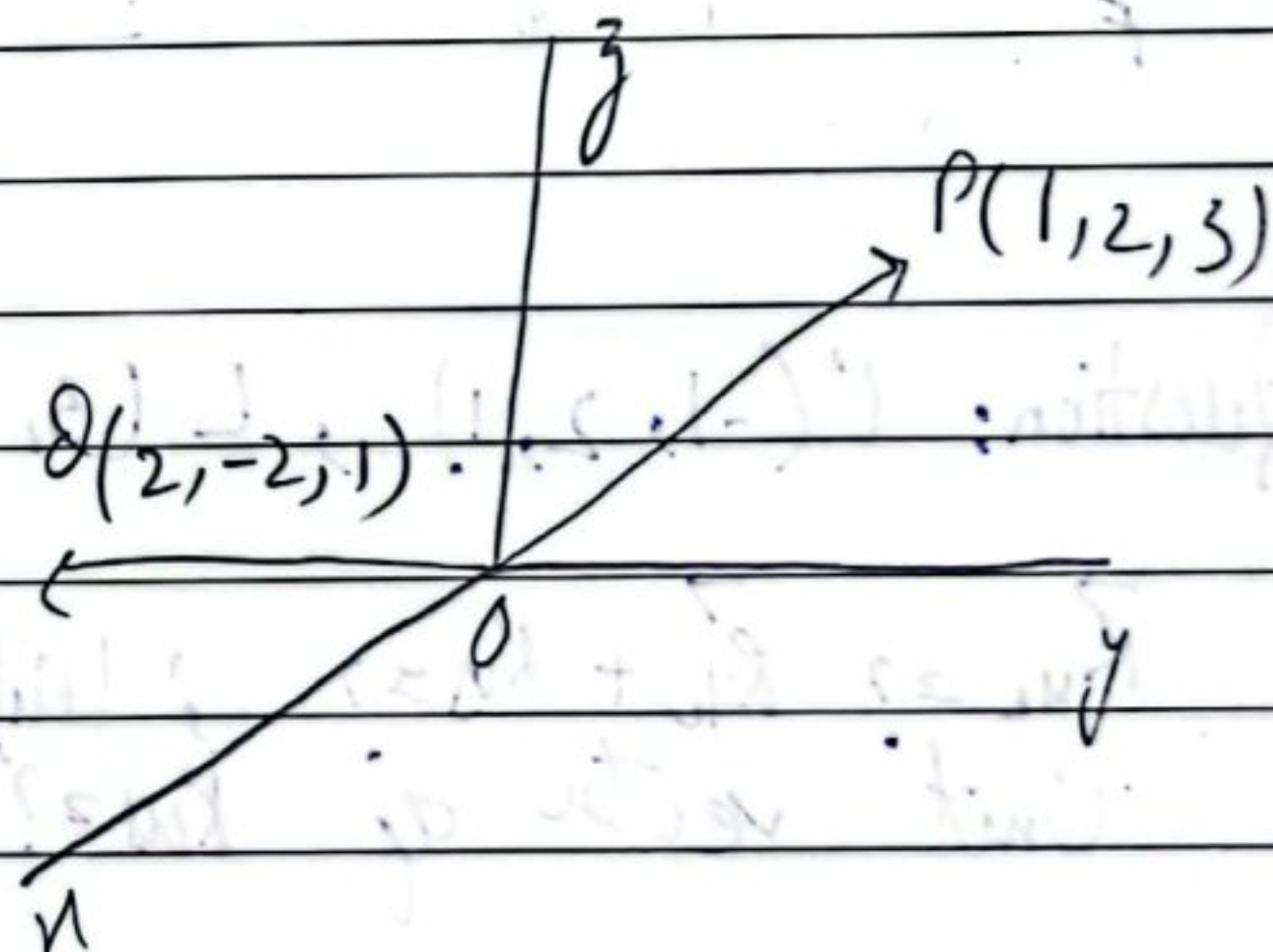
Example:-

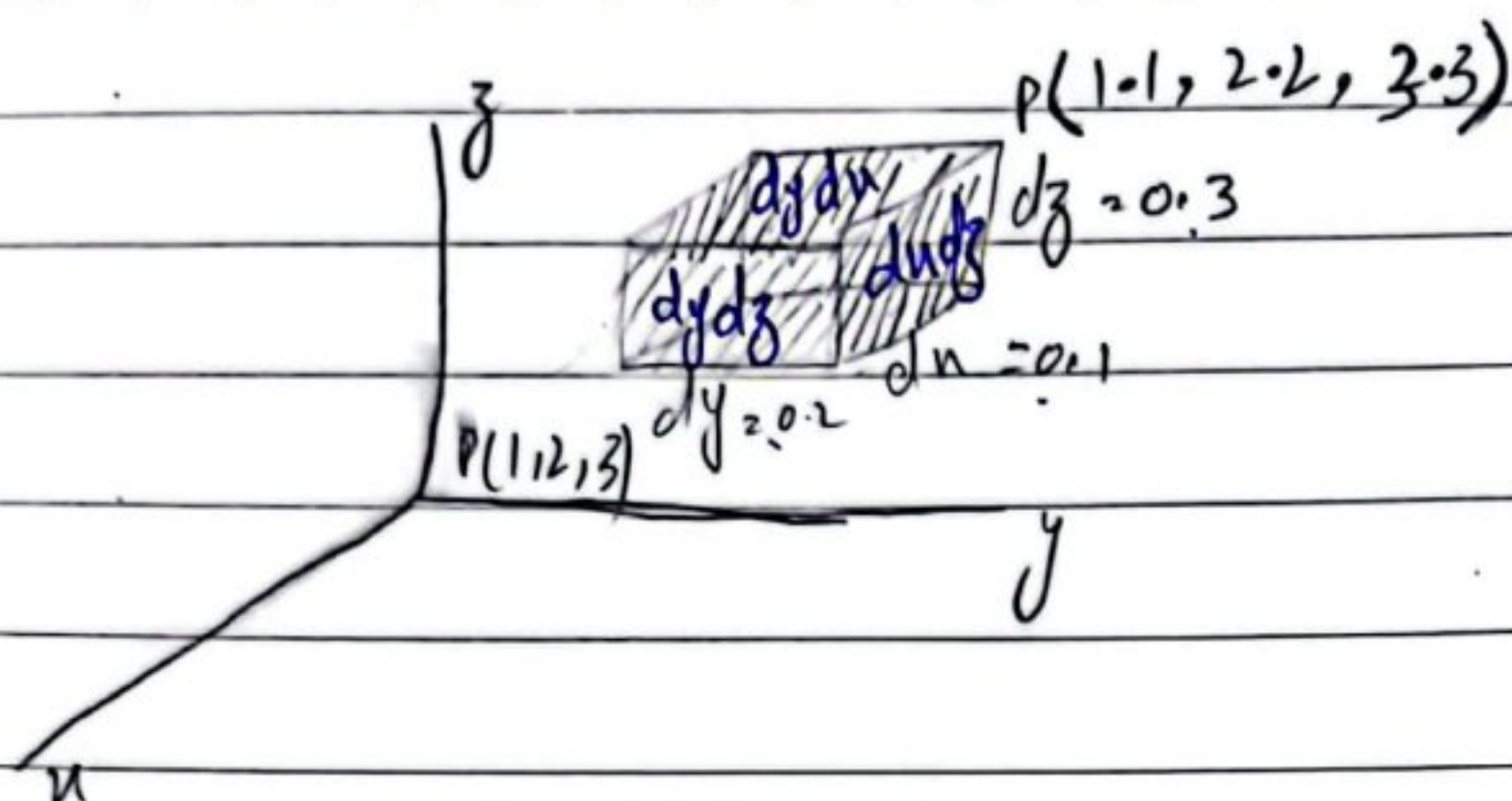
$$\vec{r}_{PO} = (1\hat{i}, -4\hat{j}, -2\hat{k})$$

$$|\vec{r}_{PO}| = \sqrt{(1)^2 + (-4)^2 + (-2)^2}$$

$$= \sqrt{1+16+4}$$

$$|\vec{r}_{PO}| = \sqrt{21}$$





$$dv = dx dy dz$$

$$|dL| = \sqrt{dx^2 + dy^2 + dz^2}$$

distance between the origin & the moved point

Question: Specify the unit vector from the origin towards the point $F(2, -2, 1)$

$$\vec{F} = (2, -2, 1)$$

$$|\vec{F}| = \sqrt{2^2 + (-2)^2 + 1^2}$$

$$= \sqrt{4 + 4 + 1}$$

$$= \sqrt{9}$$

$$|\vec{F}| = 3$$

$$\hat{F} = \frac{(2\hat{i}, -2\hat{j}, \hat{k})}{3}$$

$$\hat{F} = \left(\frac{2}{3}\hat{i} - \frac{2}{3}\hat{j} + \frac{1}{3}\hat{k} \right)$$

Question: $U(-1, 2, 1)$, $L(3, -3, 0)$, $S(-2, -3, -4)$

$$\vec{R}_{UL} = ? \quad \vec{R}_{UL} + \vec{R}_{US} = ? \quad ; \quad |\vec{R}_U| = ? \quad |\vec{R}_S| = ?$$

Unit vector of $\vec{R}_{UL} = ?$

$$R_{UL} = R_L - R_U$$

$$\vec{R}_{UL} = 4\hat{i} + (-5)\hat{j} - 1\hat{k}$$

$$R_{US} = R_S - R_U$$

$$\vec{R}_{US} = -1\hat{i} - 5\hat{j} - 5\hat{k}$$

$$\vec{R}_{UL} + \vec{R}_{US} = 3\hat{i} - 10\hat{j} - 6\hat{k}$$

$$|R_U| = \sqrt{(-1)^2 + (-2)^2 + (1)^2}$$

$$= \sqrt{1+4+1}$$

$$= \sqrt{6}$$

$$|R_S| = \sqrt{(-2)^2 + (-3)^2 + (-4)^2}$$

$$= \sqrt{4+9+16}$$

$$= \sqrt{29}$$

$$|R_{US}| = \sqrt{(-1)^2 + (-5)^2 + (-5)^2}$$

$$= \sqrt{1+25+25}$$

$$|R_{US}| = \sqrt{51}$$

$$\hat{i}_{US} = \frac{-1}{\sqrt{51}}\hat{i} - \frac{5}{\sqrt{51}}\hat{j} - \frac{5}{\sqrt{51}}\hat{k}$$

Cylindrical co-ordinate system:-

Real world Application:-

- Electromagnetics
- Fluid Dynamics
- Heat conduction in cycles
- Electromagnetic field in wires
- Flow in pipes

Rectangular coordinate system

cylindrical coordinate system

→ Define by ρ, ϕ and z

→ Define by x, y and z axis

→ ρ (radial distance from the z -axis)

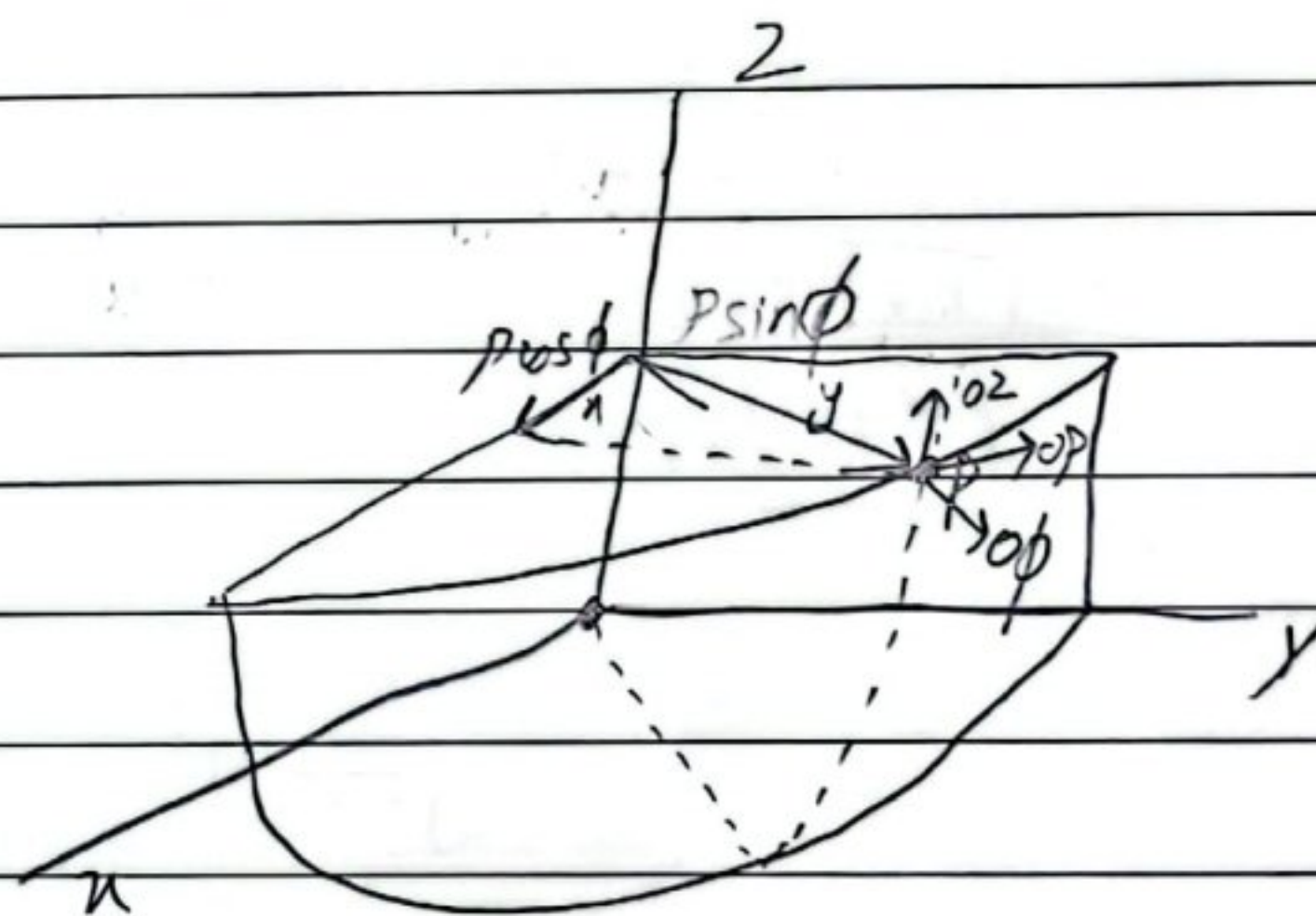
→ ϕ (Angle it makes in the xy plane)

→ z (distance along z axis)

cylindrical co-ordinates

→ surface of constant ρ are centered along z -axis

→ surface of constant ϕ are planes passing through z -axis



→ Surface of constant z are horizontal planes.

∴ ρ is always the same throughout the same cylinder.

∴ unit vector of ρ is tangent

Conversion from Rectangular co-ordinate system to cylindrical co-ordinate system.

$$\rightarrow \rho = \sqrt{x^2 + y^2}$$

→ x, y, z given

$$\rightarrow \tan \phi = \frac{y}{x} \Rightarrow \phi = \tan^{-1} \frac{y}{x}$$

$$\rightarrow \phi = \tan^{-1} \frac{y}{x}$$

$$\rightarrow z = z$$

Conversion from CCS to RCS:-

→ ρ, ϕ, z given.

$$x = \rho \cos \phi$$

$$y = \rho \sin \phi$$

$$z = z$$

Unit vectors

$$\rightarrow a_\rho, a_\phi, a_z \quad (\text{CCS})$$

$$\rightarrow a_x, a_y, a_z \quad (\text{RCS})$$

Transforming unit vector:-
Dot Product

	a_p	a_ϕ	a_z
a_u	$\cos\phi$	$-\sin\phi$	0
a_y	$\sin\phi$	$\cos\phi$	0
a_z	0	0	1

$$A = A_p a_p + A_\phi a_\phi + A_z a_z$$

$$A = A_u a_u + A_y a_y + A_z a_z$$

$$A_{rect} = A_u a_u + A_y a_y + A_z a_z$$

$$A_{cyl} = A_p a_p + A_\phi a_\phi + A_z a_z$$

→ To find any desired component of a vector, we take the dot product of the vector and a unit vector in desired component

$$A_p = A_{rect} \cdot a_p$$

$$A_\phi = A_{rect} \cdot a_\phi$$

$$A_z = A_z$$



$$A_p = A_{rect} \cdot a_p = (A_x a_x + A_y a_y + A_z a_z) \cdot a_p$$

$$A_p = A_x(a_x \cdot a_p) + A_y(a_y \cdot a_p) + A_z(a_z \cdot a_p)$$

$$A_p = A_x \cos \phi + A_y \sin \phi + A_z(0)$$

$$A_p = A_x \cos \phi + A_y \sin \phi$$

$$A_\phi = A_{rect} \cdot a_\phi = (A_x a_x + A_y a_y + A_z a_z) \cdot a_\phi$$

$$A_\phi = A_x(a_x \cdot a_\phi) + A_y(a_y \cdot a_\phi) + A_z(a_z \cdot a_\phi)$$

$$A_\phi = -A_x \sin \phi + A_y \cos \phi + A_z(0)$$

$$A_\phi = -A_x \sin \phi + A_y \cos \phi$$

$$A_z = A_z$$

$B = y_{ax} - n_{ay} + z_{az}$
 Transform it into CCS

$$B_{cyl} = B_p a_p + B_\phi a_\phi + B_z a_z$$

$$B_p = B \cdot a_p = (y_{ax} - n_{ay} + z_{az}) \cdot a_p$$

$$B_\phi = B \cdot a_\phi = (y_{ax} - n_{ay} + z_{az}) \cdot a_\phi$$



Date: ____/____/____

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$$B\rho = y \cos \phi - x \sin \phi$$

$$B\phi = -y \sin \phi - x \cos \phi$$

Assessment :-

Example # 1.3Transform $B = y a_x - x a_y + z a_z$ into cylindrical co-ordinates.

Solution:-

$$B_{cyl} = B_{\rho} a_{\rho} + B_{\phi} a_{\phi} + B_z a_z$$

$$B\rho = (y a_x - x a_y + z a_z) \cdot a_{\rho}$$

$$= y(a_x \cdot a_{\rho}) - x(a_y \cdot a_{\rho}) + z(a_z \cdot a_{\rho})$$

$$= y \cos \phi - x \sin \phi + z(0)$$

$$= y \cos \phi - x \sin \phi$$

Putting value of "y" & "x"

$$= \rho \sin \phi \cos \phi - \rho \cos \phi \sin \phi$$

$$= 0$$



$$B\phi = (y a_u - u a_y + z a_z) \cdot a\phi$$

$$= y (a_u \cdot a\phi) - u (a_y \cdot a\phi) + z (a_z \cdot a\phi)$$

$$= y (1 - \sin\phi) - u \cos\phi + z (0)$$

$$= -y \sin\phi - u \cos\phi$$

put value of "y" & "u"

$$= -p \sin^2\phi - p \cos^2\phi$$

$$= -p (\sin^2\phi + \cos^2\phi)$$

$$= -p (1)$$

$$\boxed{B\phi = -p}$$

Spherical co-ordinate system:-

RCS $\rightarrow x, y, z$

CCS $\rightarrow \rho, \phi, \theta$

SCS $\rightarrow r, \phi, \theta$

- $\rightarrow r \rightarrow$ the distance from origin to the point
- $\rightarrow$ Polar Angle $(\theta) \rightarrow$ The angle between the z-axis and the line connecting to the region.
- $\rightarrow$ Azimuthal Angle $(\phi) \rightarrow$ The angle between the x-axis and projection of the line connecting the origin to the point onto the xy-plane.

RCS $\rightarrow$ 3 planes

CCS $\rightarrow$ 2 planes, 1 curve

SCS $\rightarrow$ A sphere, cone, plane.

Unit vector:-

a_r, a_θ, a_ϕ

- $a_r \rightarrow$ Radial unit vector, directed outward from the origin, normal to the sphere
- $a_\theta \rightarrow$ tangent to the surface and pointing in the direction of increasing θ , normal to the conical plane.
- $a_\phi \rightarrow$ tangent to the sphere & cone, pointing in the direction of increasing ϕ .

Transformation from RCS to SCS
 $(x, y, z) \rightarrow (r, \theta, \phi)$

$$r = \sqrt{x^2 + y^2 + z^2} \quad (r \geq 0)$$

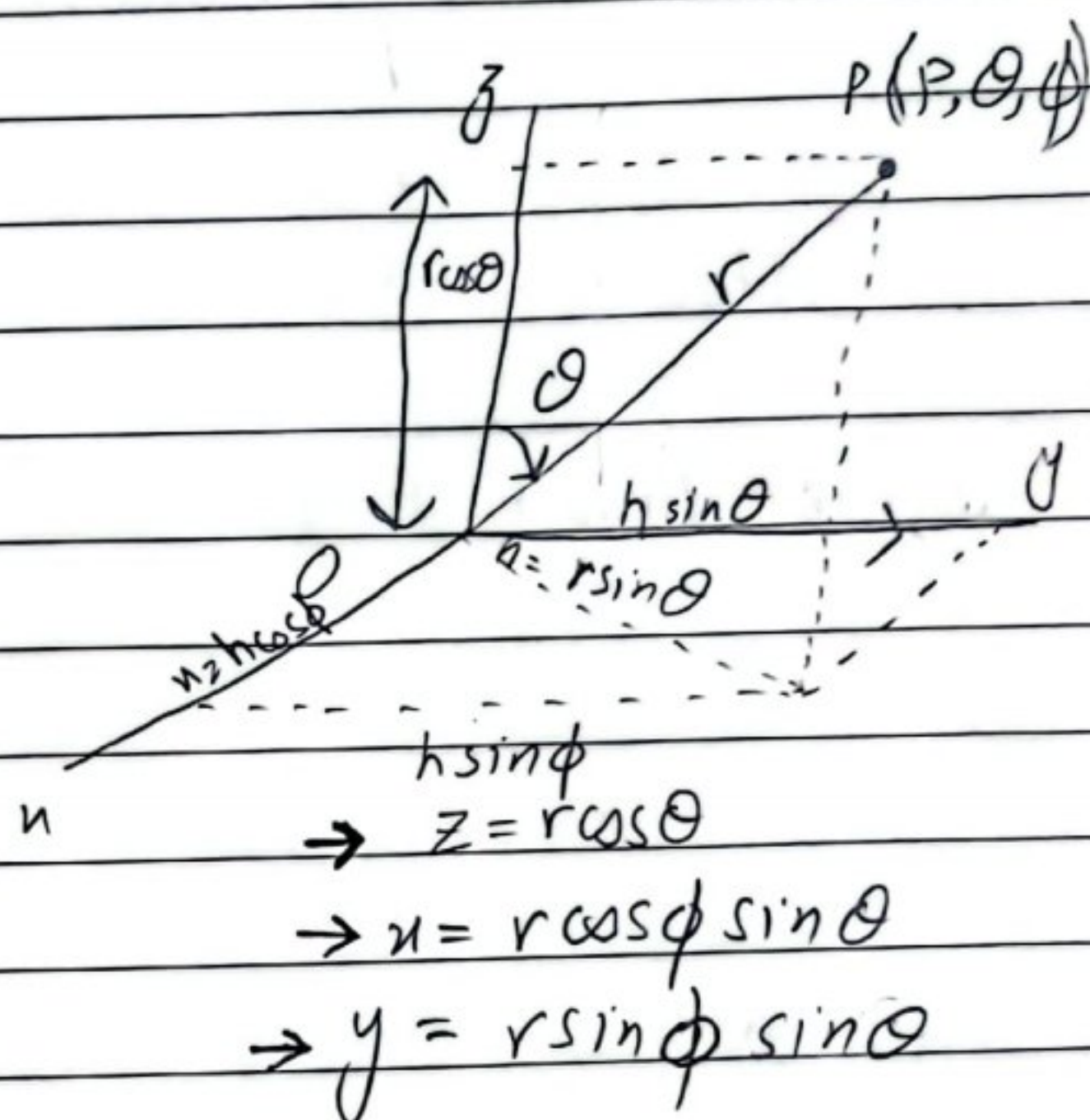
$$\phi = \tan^{-1}\left(\frac{y}{x}\right) \quad (0 \leq \phi \leq 360)$$

$$z = r \cos \theta$$

$$\cos \theta = \frac{z}{r}$$

$$\cos \theta = \frac{z}{\sqrt{x^2 + y^2 + z^2}}$$

$$\theta = \cos^{-1}\left(\frac{z}{\sqrt{x^2 + y^2 + z^2}}\right) \quad (0 \leq \theta \leq 180^\circ)$$



Transformation of vectors:-

The transformation of vectors requires determining the dot product of spherical unit vectors with rectangular unit vectors.

Purpose of dot product:-

To find the component of a vector in the direction of a new unit vector (from SCS), we take the dot product of original vector's (from RCS) with the new vectors.

Mathematical Representation:-

In spherical co-ordinates system

$$A_{\text{sph}} = A_r \hat{r} + A_\theta \hat{\theta} + A_\phi \hat{\phi}$$

The components of the vector A_{sph} along the sphere co-ordinate axis

$$A_r = A_{\text{sph}} \cdot \hat{r}$$

$$A_\theta = A_{\text{sph}} \cdot \hat{\theta}$$

$$A_\phi = A_{\text{sph}} \cdot \hat{\phi}$$

$$A_r = A_x (\hat{x} \cdot \hat{r}) + A_y (\hat{y} \cdot \hat{r}) + A_z (\hat{z} \cdot \hat{r})$$

Dot product that give the projection of the rectangular unit vectors onto the spherical unit " $\hat{r}$ ".

"Transformation of Vectors"

$$A_{\text{RCS}} = A_x a_x + A_y a_y + A_z a_z \quad , \quad A_{\text{SCS}} = A_r a_r + A_\theta a_\theta + A_\phi a_\phi$$

$$A_r = A_{\text{RCS}} \cdot a_r$$

$$= (A_x a_x + A_y a_y + A_z a_z) \cdot a_r$$

$$= A_x (a_x \cdot a_r) + A_y (a_y \cdot a_r) + A_z (a_z \cdot a_r)$$

$$= A_x (\sin \theta \cos \phi) + A_y (\sin \theta \sin \phi) + A_z (\cos \theta)$$

$\therefore a_r$: point radially outwards from the origin.

$\therefore a_\theta$: Tangent to spherical surface and pointing in the direction of increase in θ

$$A_r = A_x \cos \phi \sin \theta + A_y \sin \phi \sin \theta + A_z \cos \theta$$

$\therefore$ Tangent to the centre of constant 'r' and ' θ ' pointing the direction of increasing ϕ .

$$A_\theta = A_{\text{RCS}} \cdot a_\theta$$

$$= A_x (a_x \cdot a_\theta) + A_y (a_y \cdot a_\theta) + A_z (a_z \cdot a_\theta)$$

$$A_\theta = A_x \cos \theta \cos \phi + A_y \cos \theta \sin \phi + A_z \sin \theta$$

$$A_\phi = A_x (a_x \cdot a_\phi) + A_y (a_y \cdot a_\phi) + A_z (a_z \cdot a_\phi)$$

$$= -A_x \sin \phi + A_y \cos \phi + A_z \cdot 0$$

Question:- $G = \frac{yz}{y} a_n \rightarrow$ express it in spherical co-ordinate system.

Step 1:-

$$G = G_{r\hat{r}} + G_{\theta\hat{\theta}} + G_{\phi\hat{\phi}}$$

$$\begin{aligned} G_r = G_{r\hat{r}} &= \frac{yz}{y} a_n \cdot \hat{r} \\ &= \frac{(r \sin \theta \cos \phi)(r \cos \theta)}{r \sin \theta \sin \phi} (a_n \cdot \hat{r}) \\ &= \frac{r \cos \theta \cos \phi}{\sin \phi} (\sin \theta \cdot \cos \phi) \end{aligned}$$

$$G_r = \frac{r \cos^2 \phi \cos \theta \sin \theta}{\sin \phi}$$

Step 2:-

$$\begin{aligned} G_{\theta} = G_{\theta\hat{\theta}} &= \frac{yz}{y} a_n \cdot \hat{\theta} \\ &= \frac{(r \sin \theta \cos \phi)(r \cos \theta)}{r \sin \theta \sin \phi} (\cos \theta \cos \phi) \end{aligned}$$

$$G_{\theta} = \frac{r \cos^2 \theta \cos^2 \phi}{\sin \phi}$$



$$G_\phi = G \cdot a_\phi = \frac{u_z}{y} \cdot a_\phi$$

$$G_\phi = -r \cos \theta \cos \phi$$

$$G = \frac{r \cos \theta \cos^2 \theta \sin \theta}{\sin \phi} a_r + \frac{r \cos^2 \theta \cos^2 \phi}{\sin \phi} a_\theta - r \cos \theta \cos \phi a_\phi$$

$$G_\phi = G \cdot a_\phi = \frac{y^2}{y} \sin \theta \cdot a_\phi$$

$$G_\phi = -r \cos \theta \cos \phi$$

$$G = \frac{r \cos \theta \cos^2 \theta \sin \theta}{\sin \phi} dr + \frac{r \cos^2 \theta \cos^2 \phi}{\sin \phi} d\theta - r \cos \theta \cos \phi d\phi$$

Differential elements in ^{Day 6} the spherical coordinate system

Differential length:-

→ consider increasing r , θ and ϕ by small amounts →
 dr , $d\theta$ and $d\phi$ respectively.

- $dr \rightarrow$ along the radial direction (r)
- $r d\theta \rightarrow$ along the polar direction (θ)
- $r \sin \theta d\phi \rightarrow$ along the azimuthal direction (ϕ)

Differential surface Area:-

- Surface Area spanned by " r " and " θ " = $dr \times r d\theta = r dr d\theta$
- Surface area spanned by " r " and " ϕ " = $dr \times r \sin \theta d\phi = r \sin \theta dr d\phi$
- Surface area spanned by " θ " and " ϕ " = $r^2 \sin \theta d\theta d\phi$

• Differential Volume :-

$$= dr \times r d\theta \times r \sin\theta d\phi$$

$$= r^2 \sin\theta dr d\theta d\phi = dv$$

For total volume integrate it.

$$V = \int \frac{r^3 \sin\theta d\theta d\phi}{3} \quad \text{wrt "r"}$$

$$V = \int -\frac{r^2 \cos\theta}{3} dr d\phi \quad \text{wrt "}\theta\text{"}$$

$$V = \int \frac{r^3 \sin\theta}{3} d\theta d\phi \quad \text{wrt "}\phi\text{"}$$

$$\iiint dv = \frac{r^3}{3} \cos\theta \phi$$



$$① (-3, 2, 1)$$

$$x = -3, y = 2, z = 1$$

$$r = \sqrt{x^2 + y^2 + z^2}$$

$$r = \sqrt{(-3)^2 + (2)^2 + (1)^2}$$

$$= \sqrt{9+4+1}$$

$$r = \sqrt{14}$$

$$\phi = \tan^{-1} \frac{y}{x}$$

$$\phi = \tan^{-1} \left(\frac{2}{-3} \right)$$

$$\phi = 146.3^\circ$$

$$\theta = \cos^{-1} \left(\frac{z}{\sqrt{x^2 + y^2 + z^2}} \right)$$

$$\theta = \cos^{-1} \left(\frac{1}{\sqrt{14}} \right)$$

$$\theta = 75.037^\circ$$

$$② D = (r=5, \theta=20^\circ, \phi=-70^\circ)$$

$$x = r \sin \theta \cos \phi$$

$$x = (5) \sin(20^\circ) \cos(-70^\circ)$$

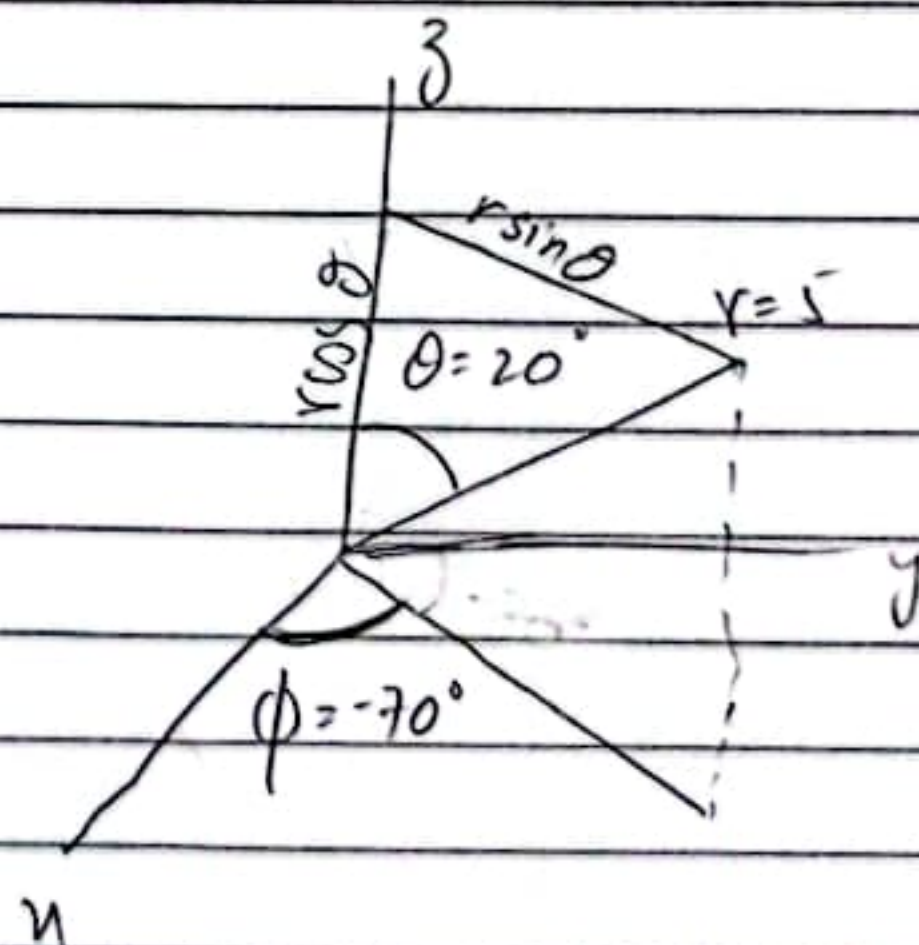
$$x = 0.584$$

$$y = r \sin \theta \sin \phi$$

$$y = -1.607$$

$$z = r \cos \theta$$

$$z = 4.698$$



Coulomb's law:-

Force between the two charged particles is: $F = k \frac{Q_1 Q_2}{R^2}$

Directly

- Proportional to the magnitude of each charge (Q_1 & Q_2)
- Inversely proportional to the square of distance b/w the two charges (R)
- Acts along the line joining the two charge.
- Is repulsive if the charge are of same sign and attractive if they are of opposite sign
- Q_1 & Q_2 are the magnitude of the charges
- R is the distance between the charges
- k is the constant of proportionality

$$k = \frac{1}{4\pi\epsilon_0}$$

- ϵ_0 is the permittivity of free space

$$\epsilon_0 = 8.854 \times 10^{-12} \text{ F/m}$$

Components of Coloumb's law:-

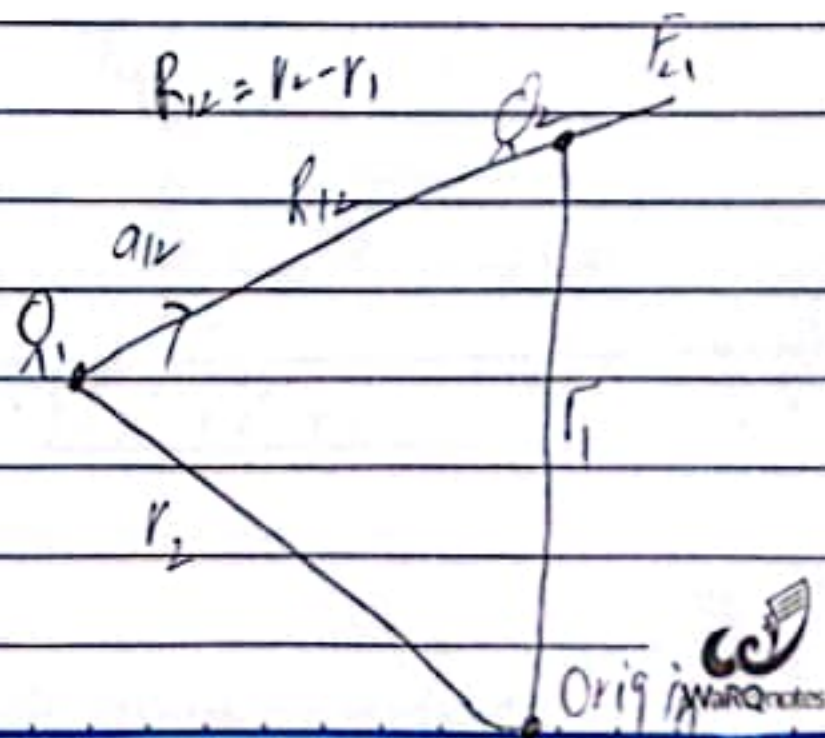
- 1- Permittivity of free space (ϵ_0) $\rightarrow$ crucial of determining how electric field interact in a vacuum. It is not dimensionless and has units $C^2/N \cdot m^2$ or F/m
- 2- The coloumb (C):- It is the SI unit of electric charge. The charge of a single electron is $1.602 \times 10^{-19} C$. 1 Coloumb corresponds to 6×10^{18} electron
- 3- Force magnitude:- Force between charges of 1 C, separated by 1m in vacuum.
 $= 9 \times 10^9 N$

Day 8

Coloumb's law (Vector form)

$$F_{12} = \frac{Q_1 Q_2}{4\pi \epsilon_0 R_{12}^2} \hat{a}_{12}$$

$$a_{12} = \frac{R_{12}}{|R_{12}|} = \frac{r_2 - r_1}{|r_2 - r_1|}$$



Electric field Intensity:-

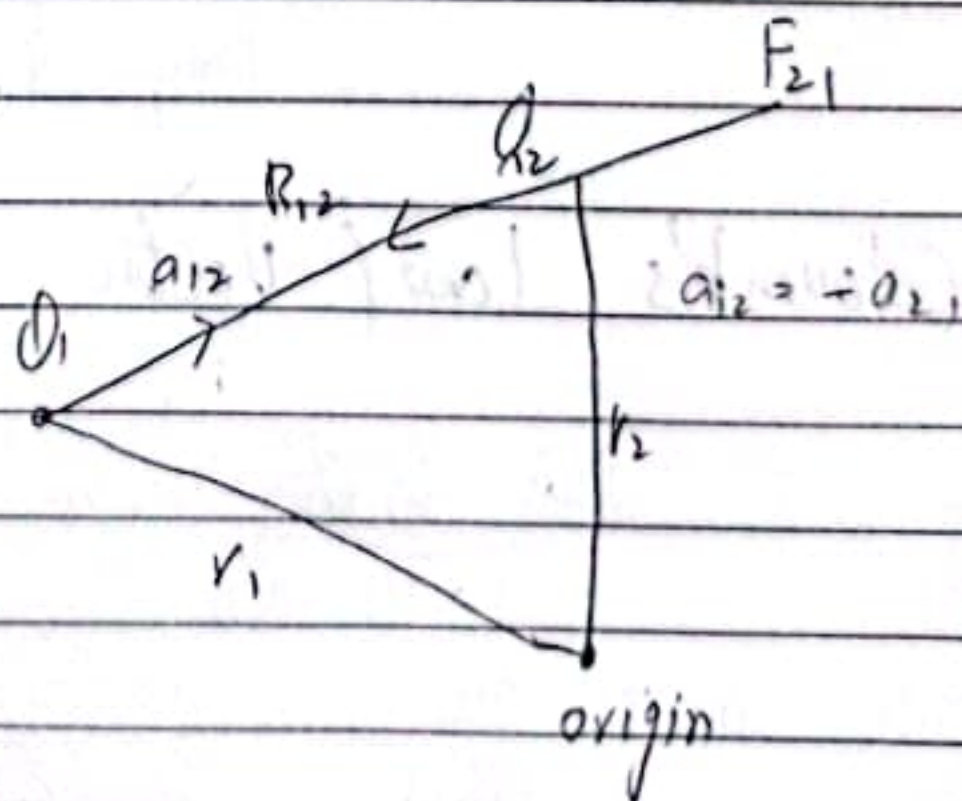
Consider a fixed charge Q_1 and the test charge Q_t is moved around Q_1

→ $\hat{a}_{1t}$ is the unit vector pointing from Q_1 to Q_t

$$F_t = \frac{Q_1 \cdot Q_t}{4\pi\epsilon_0 R_{1t}^2} \hat{a}_{1t}$$

Electric field intensity is the force experienced by a unit positive charge placed at a point in the field due to charge Q_1

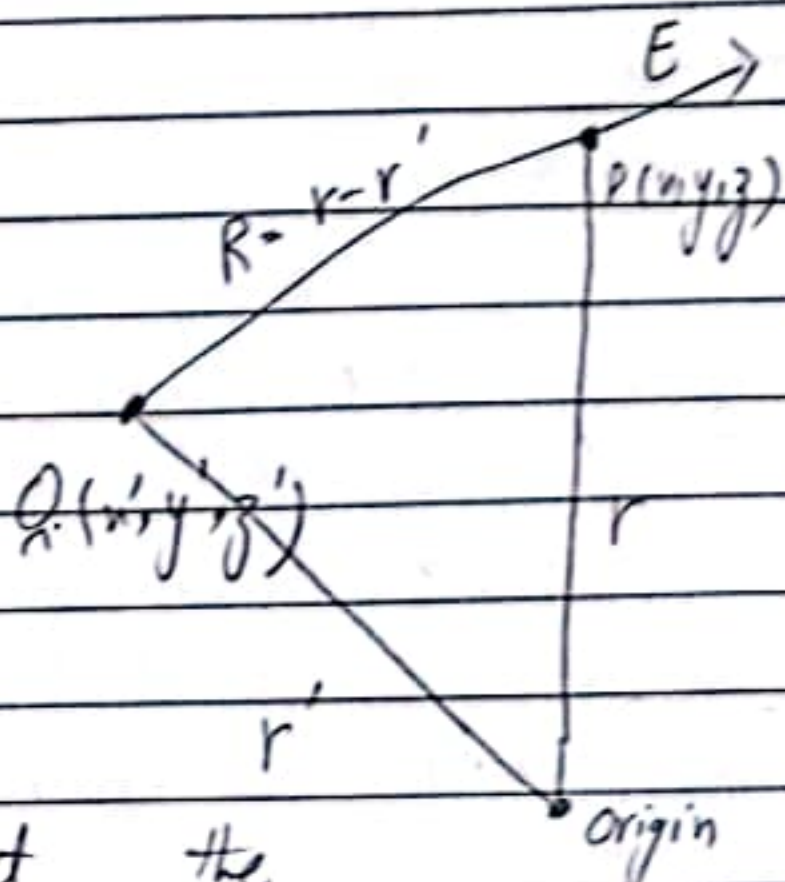
$$E_t = \frac{F_t}{Q_t} = \frac{Q_1}{4\pi\epsilon_0 R_{1t}^2} \hat{a}_{1t}$$



$$F = \frac{Q_1 Q_2}{4\pi\epsilon_0 R_{12}^2} \hat{a}_{21}$$

$$E = \frac{Q}{4\pi\epsilon_0 R^2} \hat{a}_R$$

$\hat{a}_R \rightarrow$ is the unit vector pointing from the charge Q to the point where E is measured



If the charge is placed at the origin of SCS, the unit vector $\hat{a}_R$ becomes the radial unit vector $\hat{a}_r$ and distance " R " become r .

Electric field in Rectangular Co-ordinate system:-

Suppose the charge " Q " is located at position $\vec{r}' = x'\hat{a}_x + y'\hat{a}_y + z'\hat{a}_z$

We want to find the electric field intensity at a point $\vec{r} = x\hat{a}_x + y\hat{a}_y + z\hat{a}_z$

The distance between charge and the point

of interest is $\vec{R} = \vec{r} - \vec{r}'$

magnitude of $\vec{R}$ is $|\vec{R}| = \sqrt{(x-x')^2 + (y-y')^2 + (z-z')^2}$

$$E = \frac{Q}{4\pi\epsilon_0 R^2} \hat{a}_{r-r'}$$

$$E = \frac{Q}{4\pi\epsilon_0 R^2} \left(\frac{\vec{r} - \vec{r}'}{|\vec{r} - \vec{r}'|} \right)$$

$$E = \frac{Q}{4\pi\epsilon_0 |\vec{r} - \vec{r}'|^2} \times \frac{(\vec{r} - \vec{r}')}{|\vec{r} - \vec{r}'|}$$

$$E = \frac{Q(\vec{r} - \vec{r}')}{4\pi\epsilon_0 (|\vec{r} - \vec{r}'|)^3}$$

Example:-

$$Q_1 = 4\mu C, \quad Q_2 = -3\mu C$$

$$(2, 0, 0)$$

$$(0, 2, 0)$$

calculate total E field at (1, 1, 0)

a) distance from Q_1 to (1, 1, 0)

$$|\vec{r} - \vec{r}_1| = ?$$

$$E_1 = ?$$

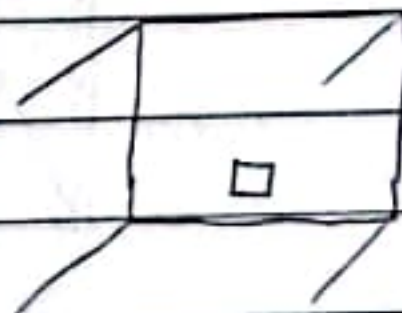
$$E_1 = \frac{4 \times 10^{-6}}{4\pi\epsilon_0 \times}$$

Day 9

Volume Charge density :-

$\rho_v \rightarrow$ describes how much charge is distributed in a small volume of space
 C/m^3 (coloumbs per cubic meter)

$$\rho_v = \frac{\text{total charge}}{\text{Total volume}} = \frac{dQ}{dV}$$



$$dQ = \rho_v dV$$

integrating b/s

$$Q = \int_V \rho_v dV = \int \int \int \rho_v dx dy dz \rightarrow \text{in RCS}$$

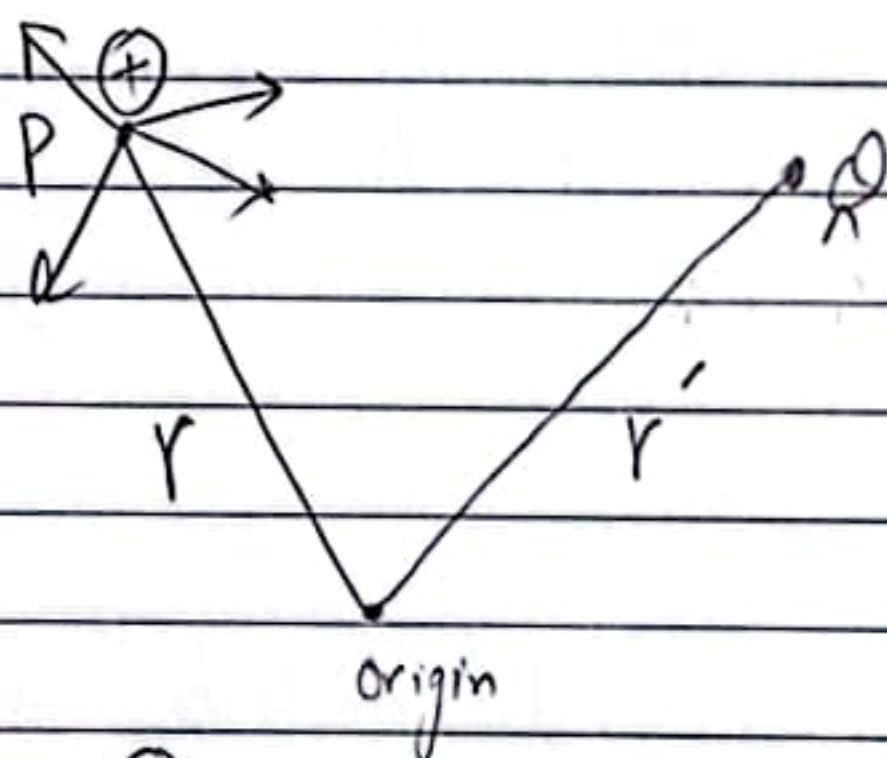
$$Q = \int \int \int \rho_v r dr d\theta dz \rightarrow \text{in CCS}$$

$$Q = \int \int \int \rho_v r^2 dr d\theta d\phi \rightarrow \text{in SCS}$$

$$dE = \frac{dQ}{4\pi\epsilon_0 |r-r'|^2} \frac{r-r'}{|r-r'|} \quad \text{--- (A)}$$

small charge point

⇒ if source point is +ve its direction is outward.



integrate (A)

$$E(r) = \int_{\text{volume}} \frac{\rho_v(r') dv}{4\pi\epsilon_0 |r-r'|^2} \frac{r-r'}{|r-r'|}$$

Total charge contained in a 2cm length of on dielectric beam

$$\rho_v = -5 \times 10^{-6} \text{ C/m}^3$$

$$Q = ?$$

$$Q = \iiint \rho_v \cdot \rho \, dz \, d\phi \, dp$$

$$Q = \int_{0.02}^{0.04} \int_0^{2\pi} \int_0^{0.01} -5 \times 10^{-6} \rho \, dp \, d\phi \, dz$$

$$= \int_{0.02}^{0.04} \int_0^{2\pi} \left[\frac{\rho^2}{2} \right]_0^{0.01} d\phi \, dz$$

$$= \int_{0.02}^{0.04} \int_0^{2\pi} \frac{(0.01^2 - 0^2)}{2} d\phi \, dz$$

$$= \int_{0.02}^{0.04} \int_0^{2\pi} 5 \times 10^{-5} d\phi \, dz$$

$$= 2.5 \times 10^{-10} \int_{0.02}^{0.04} \int_0^{2\pi} d\phi \, dz$$

$$= \left[\int_{0.02}^{0.04} [2\pi - 0] dz \right] \cdot 2.5 \times 10^{-10}$$

$$= -1.571 \times 10^{-9} \int_{0.02}^{0.04} dz$$

$$= -1.571 \times 10^{-9} (0.04 - 0.02)$$

$$Q = -3.14 \times 10^{-11}$$



$E(r) = ?$ if source point is located at
 $(0, 1, 0) = r$ and $E(r)$ needs to be found at
 $(0, 0, 1) = r'$

$$E_r = \left(\frac{-3.14 \times 10^{-11}}{4\pi \epsilon_0 |r-r'|^2} \frac{r-r'}{|r-r'|} \right)$$

$$\rho = \frac{x^2 + y^2}{z^2 + 1^2}$$

$$\rho_1 = 1$$

$$\phi = \tan^{-1} \left(\frac{y}{x} \right)$$

$$z = \tan^{-1} \left(\frac{1}{0} \right)$$

$$z = \tan^{-1}(\infty)$$

$$\phi = 90^\circ = \frac{\pi}{2}$$

$$z = 0$$

$$\rho = 0$$

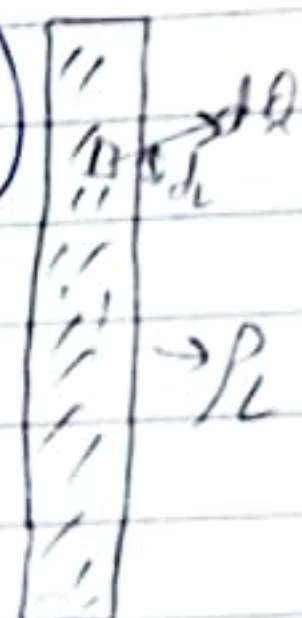
$$\phi = 0$$

$$z = 1$$



Line Charge

$$\rho_L = \frac{\text{Total Charge}}{\text{Total Length}} = \frac{dQ}{dL}$$



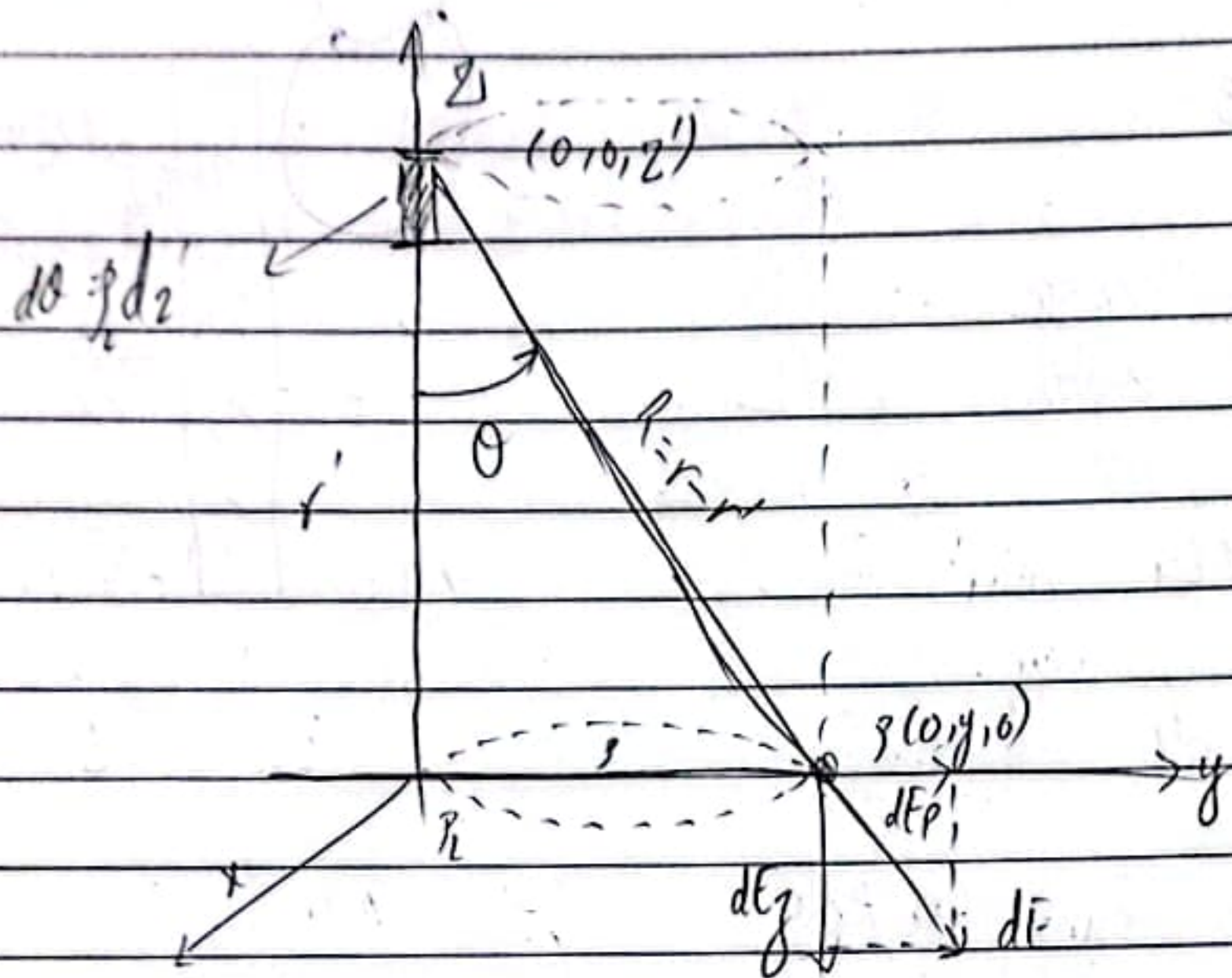
$$dQ = \rho_L dL$$

$$Q = \int_L \rho_L dL$$

$$Q = \int_x \rho_L dx \rightarrow \text{RCS}$$

$$Q = \int_p \rho_L dp \rightarrow \text{CCS}$$

$$Q = \int_r \rho_L dr \rightarrow \text{SCS}$$



$dq = \rho_L dz'$ located at a distance z' from origin

$$d\vec{E} = \frac{dq}{4\pi\epsilon_0 |\vec{r} - \vec{r}'|^3} (\vec{r} - \vec{r}')$$

$$= \frac{\rho_L dz'}{4\pi\epsilon_0 |\vec{r} - \vec{r}'|^3} (\vec{r} - \vec{r}')$$

$$\vec{r} = \rho \hat{a}_\rho, \quad \vec{r}' = z' \hat{a}_z$$

$$\vec{r} - \vec{r}' = \rho \hat{a}_\rho - z' \hat{a}_z$$

$$|\vec{r} - \vec{r}'| = \sqrt{\rho^2 + z'^2}$$

Day 10

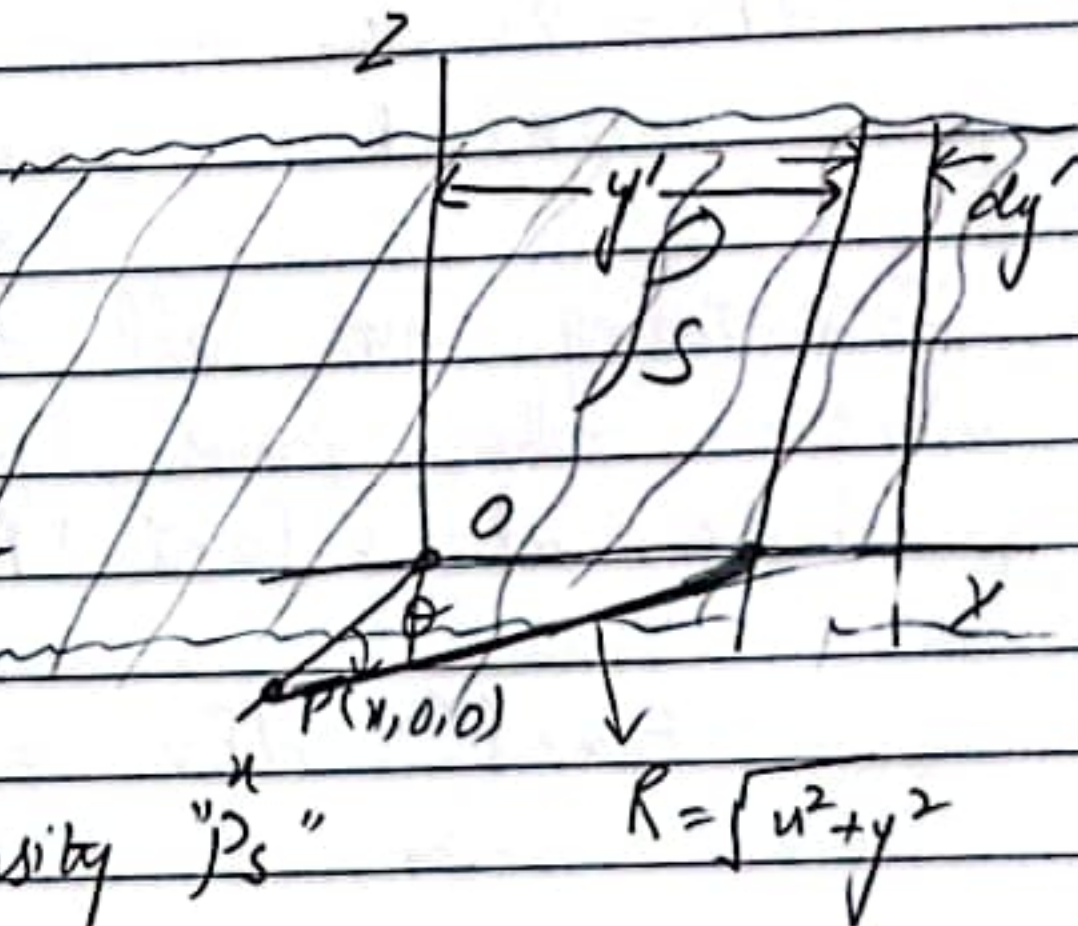
Field of a sheet charge

Surface charge density:-

Amount of electric charge per unit area of a surface (C/m^2)

Infinite sheet of charge:-

Let's place an infinite sheet of charge in the $y-z$ plane with uniform charge density " ρ_s "



→ Due to the symmetry, the electric field cannot vary along y -axis or z -axis because any point on the sheet look identical along these direction.

The line charge density of the thin strip (in a finite sheet of charge)

$$\rho_L = \rho_s dy'$$

$$R = \sqrt{x^2 + y'^2}$$

Using colomb's law, the differential contribution to the electric field

dE_u from the strip is

$$dE_u = \frac{\rho_L u}{2\pi\epsilon_0 r^2} = \frac{\rho_s dy' u}{2\pi\epsilon_0 (u^2 + y'^2)}$$

Integrating over all the differential strips along the y-axis gives the total field at E_u at Point 'P'

$$E_u = \int_{-\infty}^{\infty} dE_u = \int_{-\infty}^{\infty} \frac{\rho_s dy' u}{2\pi\epsilon_0 (u^2 + y'^2)}$$

$$= \frac{\rho_s u}{2\pi\epsilon_0} \int_{-\infty}^{\infty} \frac{dy'}{(u^2 + y'^2)} \Rightarrow \frac{\rho_s u}{2\pi\epsilon_0} \frac{1}{u} \left[\frac{y'}{u^2 + y'^2} dy' \right]$$

$$= \frac{\rho_s}{2\pi\epsilon_0} \left| \tan^{-1}\left(\frac{y}{u}\right) \right|_{-\infty}^{\infty}$$

$$= \frac{\rho_s}{2\pi\epsilon_0} \left(\tan^{-1}\left(\frac{\infty}{u}\right) - \left(\tan^{-1}\left(\frac{-\infty}{u}\right) \right) \right)$$

$$= \frac{\rho_s}{2\pi\epsilon_0} \left(\frac{\pi}{2} - \left(-\frac{\pi}{2} \right) \right)$$

$$= \frac{\rho_s}{2\pi\epsilon_0} \left(\frac{\pi}{2} \right)$$

$$\boxed{E_n = \frac{\rho_s}{2\epsilon_0} \hat{a}_n} \rightarrow \text{Finite sheet charge}$$

$$\boxed{E = \frac{Q}{4\pi\epsilon_0 R^2} \hat{a}_R} \rightarrow \text{Point charge}$$

$$\boxed{E = \frac{Q}{2\pi\epsilon_0 R} \hat{a}_R} \rightarrow \text{Line charge}$$

Day 12

Chapter #3

Electric Flux Density:-

Electric Flux

Electric flux (Ψ) is directly proportional to charge on the inner sphere. It is also called displacement flux.

$$\Psi = Q$$

Unit:-

The unit of electric flux is Coulomb

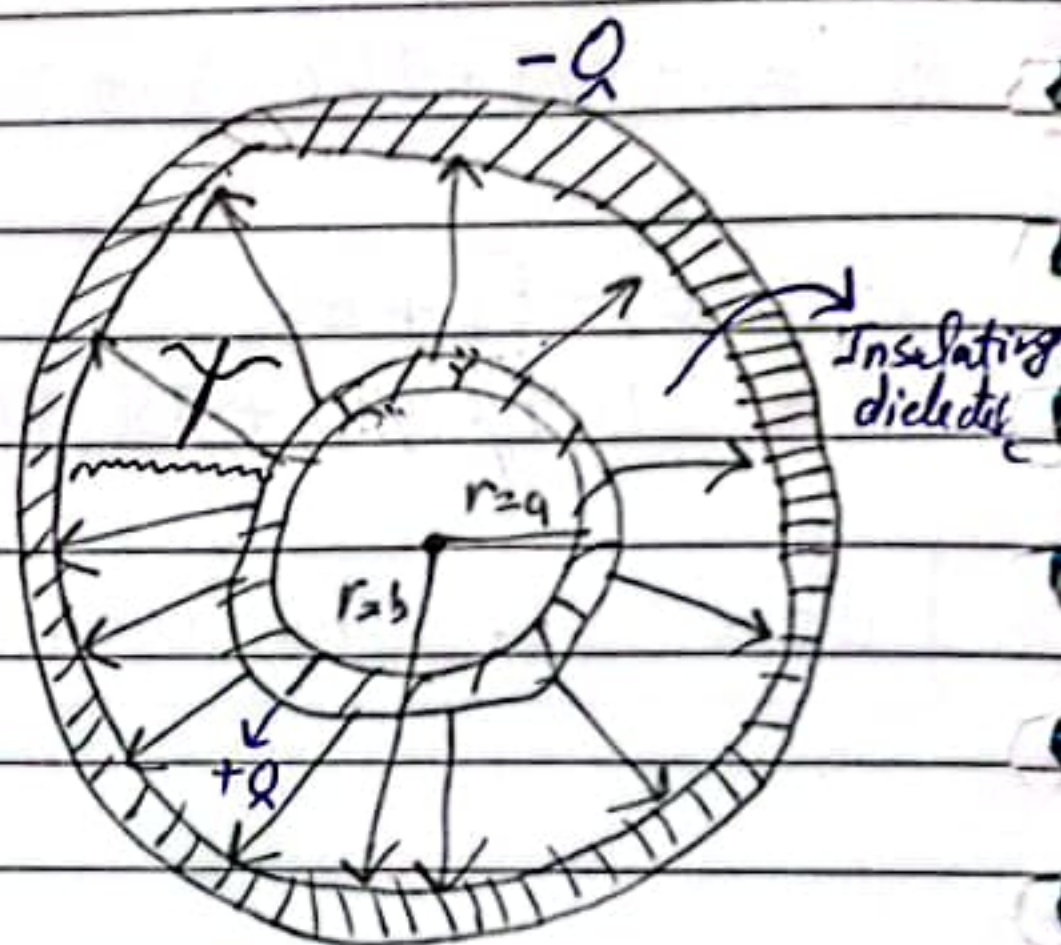
Electric Flux Density:-

Electric Flux Density is defined as the electric flux per unit area (C/m^2)

→ Electric Flux at any given point is given by the no. of flux lines crossing a surface normal to the lines divided by the surface

→ Due to the point charge, electric flux density

$$D = \frac{\Psi}{4\pi r^2} \hat{a}_r \Rightarrow \frac{Q}{4\pi r^2} \hat{a}_r$$

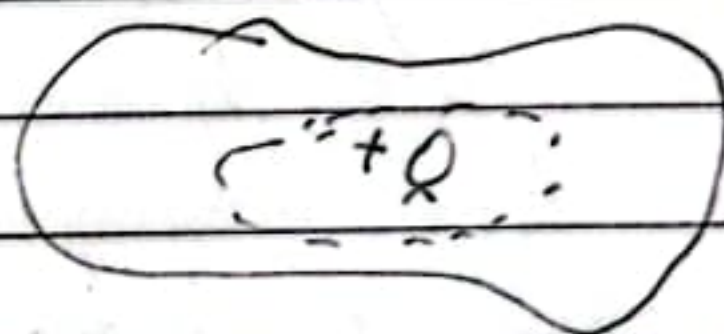


$E = \frac{Q}{4\pi\epsilon_0 r^2}$; $D = E\epsilon_0$ → for free space
 electric flux density due to point charge.

$D = E\epsilon_0\epsilon_r$ → for air

volume charge / surface charge

$E = \int_{vol} \frac{\rho_v dv}{4\pi\epsilon_0 r^2} \hat{a}_r$



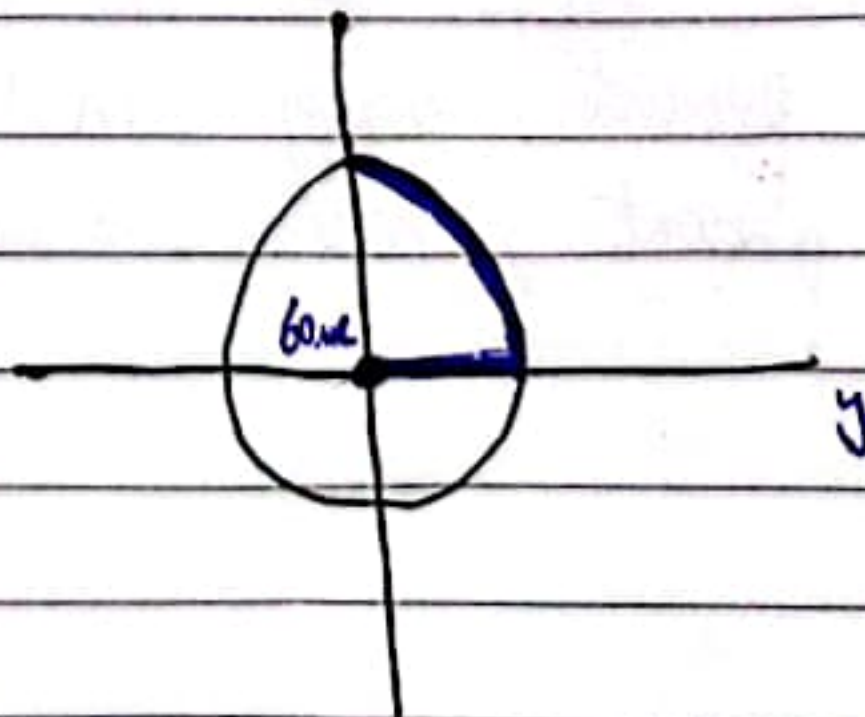
$E\epsilon_0 = \int_{vol} \frac{\rho_v dv}{4\pi r^2} \hat{a}_r \Rightarrow D = \int_{vol} \frac{\rho_v dv}{4\pi r^2} \hat{a}_r$

Question: - Given a $60\mu C$ point charge located at the origin, find the total electric flux passing through: (a) The portion of the sphere $r = 26\text{ cm}$ bounded by $0 < \theta < \frac{\pi}{2}$ and $0 < \phi < \frac{\pi}{2}$

$\Psi = 0$

$\Psi = \frac{1}{8} \times 60\mu C$

cause according to the given limits the given part is equal to the $\frac{1}{8}$ part of the sphere.



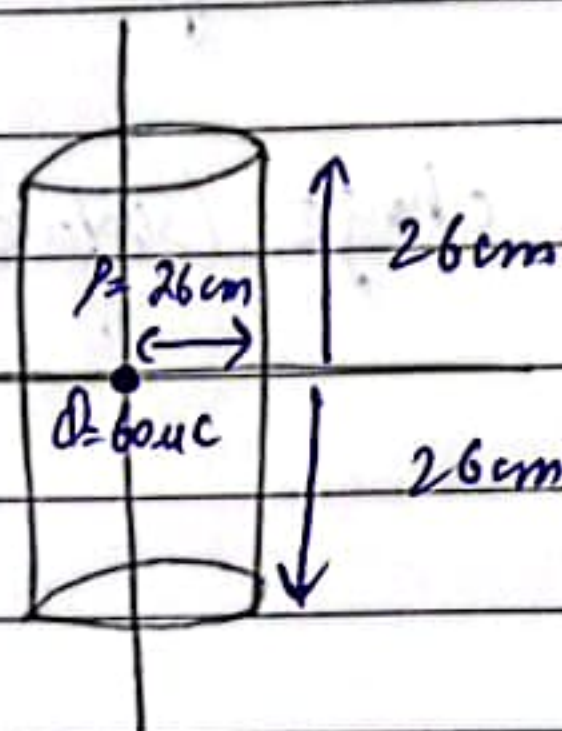
$Q = 7.5 \mu C$

b) Closed surface defined $\rho = 26 \text{ cm}$, $z = \pm 26 \text{ cm}$

Solution:-

$Q = 60 \mu C$

same charge is in closed surface. If we ask electric flux density in a particular portion then it change.

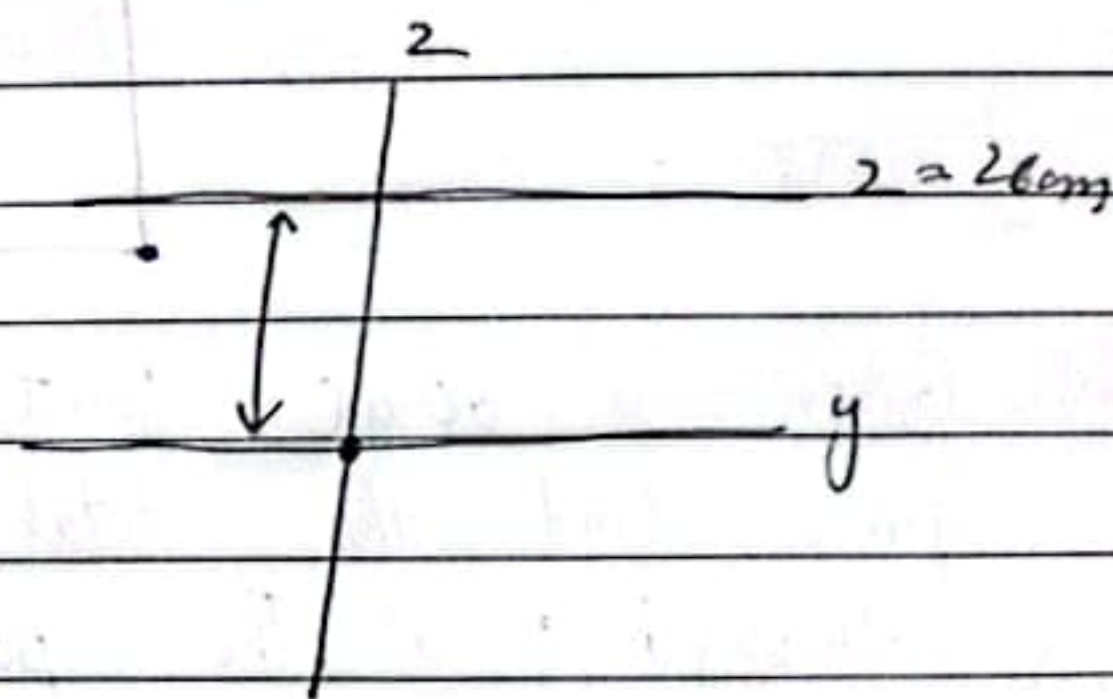


c) Plane $z = 26 \text{ cm}$

$Q = \frac{1}{2} \times 60 \mu C$

$Q = 30 \mu C$

$\frac{1}{2}$ same in the negative



co-ordinate there is no plane so electric flux density is present only in upper part in which plane is present at $z = 26 \text{ cm}$.

Question: Calculate D in RCS at Point $(2, -3, 6) \rightarrow r_1$
 a) Due to point charge $Q_1 = 55 \mu C$ at $Q(-2, 3, -6) \rightarrow r_2$
 $D = \frac{Q}{4\pi r^2} \hat{r}$

$$|r| = \sqrt{(2 - (-2))^2 + (-3 - 3)^2 + (6 - (-6))^2}$$

$$= \sqrt{(4)^2 + (-6)^2 + (12)^2}$$

$$|r| = 14 \text{ m}$$

$$\vec{r} = 4a_x - 6a_y + 12a_z$$

$$\hat{r} = \frac{2}{7} a_x - \frac{3}{7} a_y + \frac{6}{7} a_z$$

$$D = 6.37 a_x - 9.56 a_y + 19.12 a_z \quad \mu C/m^2$$

Question: Calculate D in RCS at Point $(2, -3, 6) \rightarrow r_1$
 a) Due to point charge $Q_1 = 55 \text{ mC}$ at $Q(-2, 3, -6) \rightarrow r_2$
 $D = \frac{Q}{4\pi r^2} \hat{r}$

$$|r| = \sqrt{(2 - (-2))^2 + (-3 - 3)^2 + (6 - (-6))^2}$$

$$= \sqrt{(4)^2 + (-6)^2 + (12)^2}$$

$$|r| = 14 \text{ m}$$

$$\vec{r}_2 = 4a_x - 6a_y + 12a_z$$

$$\hat{r} = \frac{2}{7} a_x - \frac{3}{7} a_y + \frac{6}{7} a_z$$

$$D = 6.37 a_x - 9.56 a_y + 19.12 a_z \quad \mu\text{C/m}^2$$

Day 13

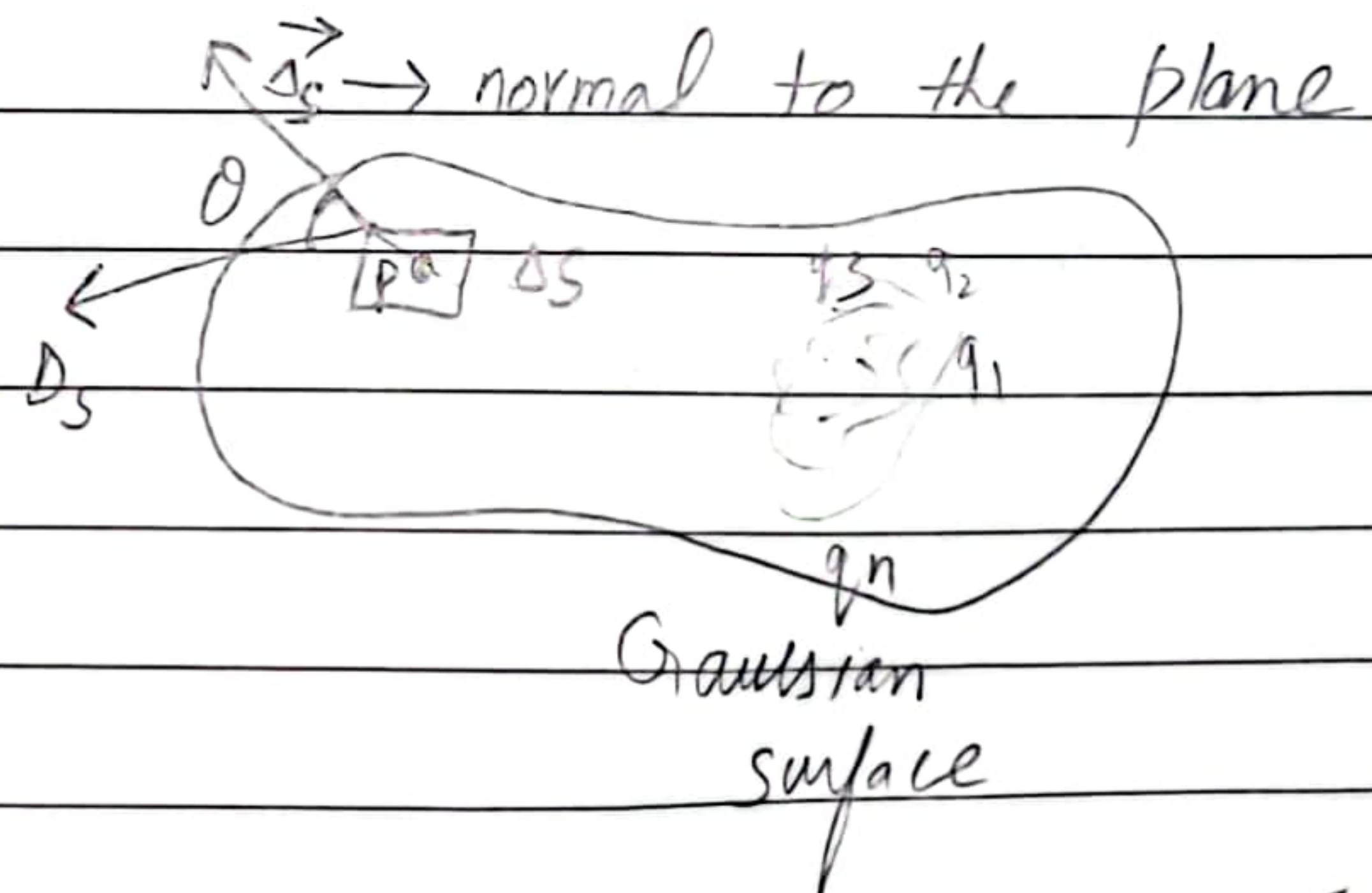
Gauss's Law:-

"The electric flux passing through any closed surface to the total charge enclosed by that surface."

$$\Delta \Psi = D_s \cdot \Delta S \quad (\text{flux passing through } \Delta S)$$

$$= D_s \Delta S \cos \theta$$

$$= D_s \cdot \Delta S$$





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$$\text{Total flux} = \Psi = \int d\Psi = \int \mathbf{D}_s \cdot d\mathbf{s}$$

$$= \oint_{\text{closed surface}} \mathbf{D}_s \cdot d\mathbf{s} \rightarrow \text{closed surface integral}$$

RCS:-

$$\Psi = Q = \iint_{xy} D_{n,y} \, dx \, dy$$

CCS:-

$$\Psi = Q = \iint_{\phi} D_{\rho,\phi} \, \rho \, d\rho \, d\phi$$

SCS:-

$$\Psi = Q = \iint_{\phi\theta} D_{\theta,\phi} \, r^2 \sin\theta \, d\theta \, d\phi$$

Point
charge

$$\Psi = Q = q_1 + q_2 + q_3 + \dots + q_n$$

$$= \epsilon Q_n = \oint \mathbf{D}_s \cdot d\mathbf{s}$$

Line
charge

$$\Psi = Q = \int \mathbf{E} \cdot d\mathbf{l} = \oint \mathbf{D}_s \cdot d\mathbf{s}$$

Surface
charge

$$\Psi = Q = \int \mathbf{E} \cdot d\mathbf{s} = \oint \mathbf{D}_s \cdot d\mathbf{s}$$



Volume charge distribution $= Q = \int_{Vol} \rho_v dv = \int_S D \cdot ds$

Problem:-

Given:

$D = 0.3 r^2 a_r \text{ nC/m}^2$ in free space

(a) Find E at Point $P(r=2, \theta=25^\circ, \phi=90^\circ)$

(b) Find the total charge within space

$r=3$

(c) Find the total electric flux leaving sphere

$r=4$

$E = \frac{Q}{4\pi\epsilon_0 r^2} a_r$

$D = E \epsilon_0$

$D = \frac{Q}{4\pi r^2} a_r$

$0.3 r^2 a_r = \frac{Q}{4\pi r^2} a_r$

$0.3 (2)^2 \times 4\pi \times (2)^2 = Q$

$Q = 6.03 \times 10^{-8} \text{ coulomb}$

(a) $D = E \epsilon_0$

$E = \frac{D}{\epsilon_0}$

$= \frac{0.3 r^2 a_r}{8.85 \times 10^{-12}}$

$= \frac{(0.3)(4)}{8.85 \times 10^{-12}} a_r$

$D = 135.529 a_r$

Example 3.1

Day 14

Application of Gauss's law: Symmetrical charge distributions.

→ only valid applicable for symmetrical surface.

$$\oint \mathbf{D} \cdot d\mathbf{s} = Q$$

Symmetry (Spherical, cylindrical, planar symmetrical)

Key condition for applying Gauss's law:

To effectively use

Gauss's law, we must choose a Gaussian surface that satisfies the following

→ $\mathbf{D}$ (Electric flux density) must be either normal or tangential to the surface

If $\mathbf{D}$ is normal to the surface
 $\mathbf{D} \cdot d\mathbf{s} = D ds$

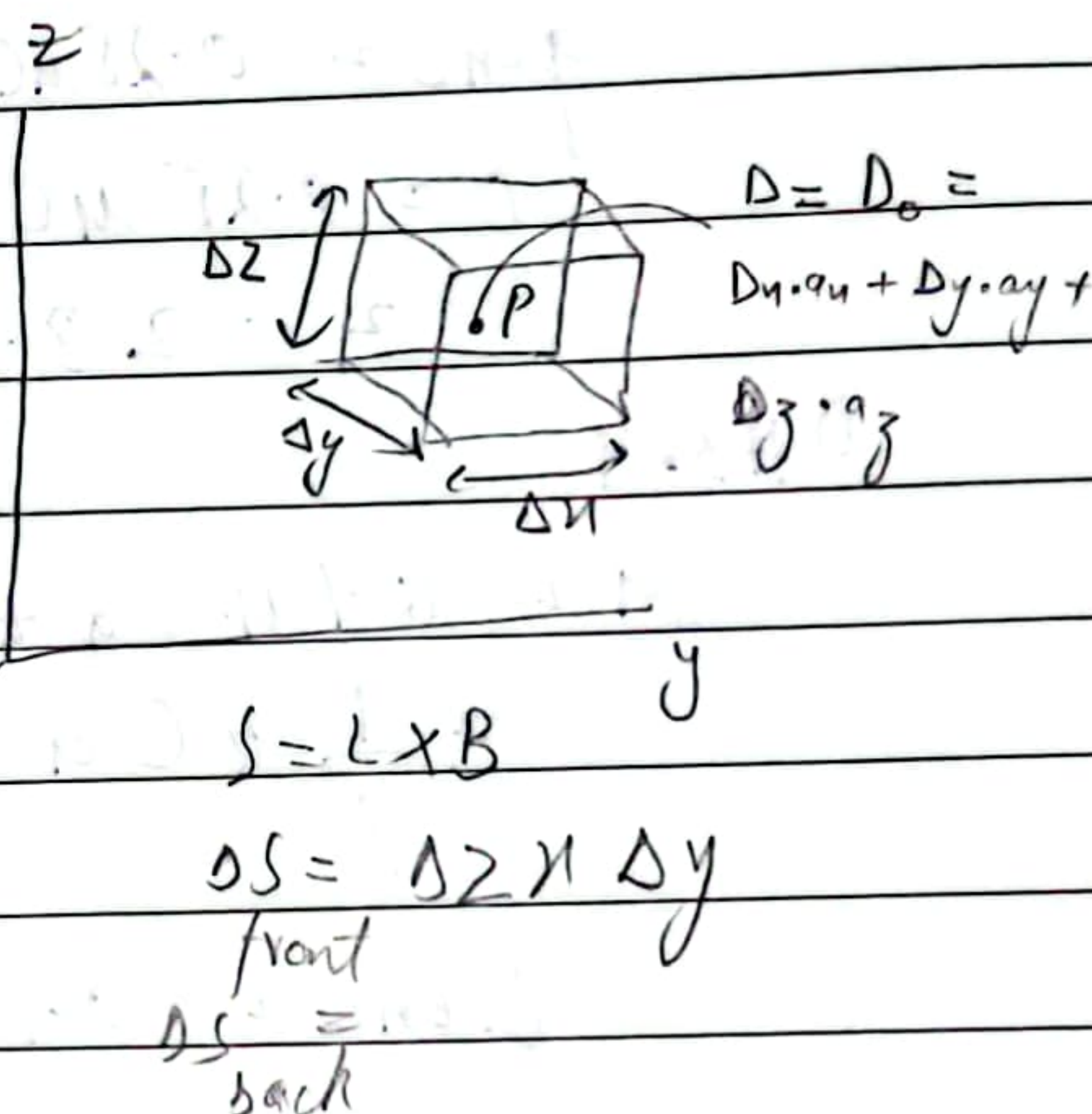
If $\mathbf{D}$ is tangential to the surface
 $\mathbf{D} \cdot d\mathbf{s} = 0$

→ The magnitude of $\mathbf{D}$ must be constant over the surface where it is normal.

Day 15

Application of Gauss's law: Differential volume element

For symmetrical surface, electric field or electric flux is constant (Gaussian surface) what if the surface is not symmetrical? $\rightarrow$ We shall consider a very small differential volume element so that Gauss's law can be applied on it to determine electric flux.



Gauss's law: Front and Back surface

$$\oint D \cdot ds = \int_{\text{front}} D \cdot ds + \int_{\text{Back}} D \cdot ds + \int_{\text{Right}} D \cdot ds + \int_{\text{Left}} D \cdot ds + \int_{\text{Top}} D \cdot ds + \int_{\text{bottom}} D \cdot ds$$

$$\int_{\text{front}} D \cdot ds = D_{\text{front}} \cdot \Delta S_{\text{front}} = D_{\text{front}} \cdot \Delta y \cdot \Delta z \cdot \Delta x$$

Now to find D_{front} (D_x at the front surface)

The front face is at a distance of $\frac{\Delta x}{2}$ from point P :

$$D_{\text{front}} = D_{x0} + \frac{\Delta x}{2} \times \text{Rate of change in } x$$

Problem # 3.6, 3.7, 3.8, 3.9

Exp # 3.3, 3.4, 3.5

Divergence and Maxwell's Equation

$$\frac{\partial(D_x)}{\partial x} + \frac{\partial(D_y)}{\partial y} + \frac{\partial(D_z)}{\partial z} = \lim_{\Delta V \rightarrow 0} \oint \frac{D \cdot ds}{\Delta V}$$

$$= \lim_{\Delta V \rightarrow 0} \frac{Q_{enc}}{\Delta V} = \rho_v$$

$$\frac{\partial(A_x)}{\partial x} + \frac{\partial(A_y)}{\partial y} + \frac{\partial(A_z)}{\partial z} = \lim_{\Delta V \rightarrow 0} \int \frac{A \cdot ds}{\Delta V}$$

A could be velocity, force or any vector quantity

$$\boxed{\text{Divergence of } A = \text{div } A = \lim_{\Delta V \rightarrow 0} \frac{\oint A \cdot ds}{\Delta V}} \rightarrow \text{Maxwell's Equation}$$

SCS $\rightarrow$ Spherical co-ordinate system

$$\text{div } D = \frac{1}{r^2} \frac{\partial}{\partial r} (r^2 D_r) + \frac{1}{r \sin \theta} \frac{\partial}{\partial \theta} (\sin \theta D_\theta) +$$

$$\frac{1}{r \sin \theta} \frac{\partial D_\phi}{\partial \phi}$$



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→ $\int_S \mathbf{D} \cdot d\mathbf{s}$ represent the flux through the surface S enclosing the volume (double integral)

→ $\int_{vol} (\nabla \cdot \mathbf{D}) dv$ is the integral of divergence Δ through out the volume. (triple integral)

Divergence of a vector fields

The divergence theorem states that
"The total flux exiting a closed surface is equal to the integral of divergence of the field within the volume enclosed by the surface."

Divergence in different co-ordinate systems:
In cylindrical co-ordinate system:

$$\nabla \cdot \mathbf{D} = \frac{1}{\rho} \frac{\partial}{\partial \rho} (\rho D_\rho) + \frac{1}{\rho} \frac{\partial D_\phi}{\partial \phi} + \frac{\partial D_z}{\partial z}$$



Divergence of a vector field:

Divergence represent the rate at which the vector field is "spreading out" from a point.

If we take

$$\mathbf{D} = D_x \hat{a}_x + D_y \hat{a}_y + D_z \hat{a}_z$$

then

$$\nabla \cdot \mathbf{D} = \frac{\partial (D_x)}{\partial x} + \frac{\partial (D_y)}{\partial y} + \frac{\partial (D_z)}{\partial z}$$

The divergence of a vector field at any point represents that whether the field is acting as a source (positive divergence) or as a sink (negative divergence $\rightarrow$ converging) at that point.

Divergence theorem connects the behaviour of a vector field inside a volume to the behaviour of the field on the surface enclosing that volume. Mathematically,

$$\oint_S \mathbf{D} \cdot d\mathbf{s} = \int_{vol} (\nabla \cdot \mathbf{D}) dv$$

from Gauss's law

$$\oint_S \mathbf{D} \cdot d\mathbf{s} = Q_{enc} = \int_{vol} (\nabla \cdot \mathbf{D}) dv \quad \rightarrow \rho_v = \nabla \cdot \mathbf{D}$$

Day 16

Divergence Theorem:

The Del theorem (∇) is a vector operator used to perform several important operations in vector calculus, including divergence, gradient and curl.

$$\nabla = \frac{\partial}{\partial x} \hat{a}_x + \frac{\partial}{\partial y} \hat{a}_y + \frac{\partial}{\partial z} \hat{a}_z$$

$\hat{a}_x$, $\hat{a}_y$, $\hat{a}_z$ are unit vector in the x , y and z direction. Partial derivatives indicate rates of change in the respective direction.

The del operator behaves like a vector but instead of returning scalars, it returns partial derivatives when applied to scalar, it returns partial derivatives when applied to scalar and vector field.

- 1- Gradient (∇u) $\rightarrow$ applied to scalar field 'u' yield a vector
- 2- Divergence ($\nabla \cdot D$) $\rightarrow$ applied to a vector field $\rightarrow$ yield a scalar
- 3- Curl ($\nabla \times D$) $\rightarrow$ Applied to a vector $\rightarrow$ yield another vector

$$\oint_{\text{Total}} \mathbf{D} \cdot d\mathbf{s} = \underbrace{Q_{\text{enclosed}}}_{\text{divergence}}$$

$$Q_{\text{enclosed}} = \left(\frac{\partial D_x}{\partial x} + \frac{\partial D_y}{\partial y} + \frac{\partial D_z}{\partial z} \right) \Delta x \Delta y \Delta z$$

$$Q_{\text{enclosed}} = \nabla \cdot \mathbf{D} \cdot \Delta V$$

$$\frac{Q_{\text{enclosed}}}{\Delta V} = \nabla \cdot \mathbf{D}$$

$$\boxed{\nabla \cdot \mathbf{D} = \rho_v}$$

(Example 3.3)

Now along u-axis combine (adding both eq.)

$$D_{\text{front}} + D_{\text{back}} = \left(\frac{D_{u0} + \frac{\Delta u}{2} \frac{\partial D_u}{\partial u} \right) \Delta y \Delta z - \left(\frac{D_{u0} + \frac{\Delta u}{2} \frac{\partial D_u}{\partial u} \right) \Delta y \Delta z$$

$$2 D_{u0} \Delta y \Delta z$$

$$D_{\text{front}} + D_{\text{back}} = \left(\frac{\partial D_u}{\partial u} + \frac{\Delta u}{2} \frac{\partial^2 D_u}{\partial u^2} \right) \Delta y \Delta z - \left(\frac{\partial D_u}{\partial u} + \frac{\Delta u}{2} \frac{\partial^2 D_u}{\partial u^2} \right) \Delta y \Delta z$$

$$\text{along u-axis} = \frac{\partial D_u}{\partial u} \Delta u \Delta y \Delta z$$

Along y-axis:

$$\oint_{y\text{-axis}} D \cdot ds = \frac{\partial D_y}{\partial y} \Delta u \Delta y \Delta z$$

Along z-axis:

$$\oint_{z\text{-axis}} D \cdot ds = \frac{\partial D_z}{\partial z} \Delta u \Delta y \Delta z$$

$$= D_{n_0} + \frac{\Delta n}{2} \cdot \frac{\partial D_n}{\partial n}$$

where D_{n_0} is value of D_n at P and Partial derivate is used to express the rate of change of D_n with n

$$D_{\text{front}} = \left(D_{n_0} + \frac{\Delta n}{2} \frac{\partial D_n}{\partial n} \right) \Delta y \Delta z$$

Now to find D_{back} (D_n at the back surface)

The ~~front~~ back face is at a distance of $-\frac{\Delta n}{2}$ from point P

$$\int_{\text{back}} D \cdot d\mathbf{s} = D_{\text{back}} \cdot \Delta S_{\text{back}} = D_{\text{back}} \cdot (\Delta y \Delta z (-\mathbf{a}_n))$$

$$= -D_{\text{back}} \Delta y \Delta z \quad \text{--- (2)}$$

$$D_{\text{back}} = D_{n_0} - \frac{\Delta n}{2} \frac{\partial D_n}{\partial n}$$

put in (2)

$$\begin{aligned} \int_{\text{back}} D \cdot d\mathbf{s} &= - \left(D_{n_0} - \frac{\Delta n}{2} \frac{\partial D_n}{\partial n} \right) \Delta y \Delta z \\ &= \left(-D_{n_0} + \frac{\Delta n}{2} \frac{\partial D_n}{\partial n} \right) \Delta y \Delta z \end{aligned}$$

Coaxial cable with cylindrical symmetry:-

$$D_s = D_p = \frac{Q}{2\pi PL} \Rightarrow \boxed{Q = D_s 2\pi PL} \quad P_s: \text{surface charge density}$$

The total charge on a length L of the inner conductor is

$$Q = \int_{z=0}^L \int_{\phi=0}^{2\pi} P_s a \, d\phi \, dz = \boxed{2\pi a L P_s}$$

$$D_s 2\pi PL = 2\pi a L P_s = Q$$

$$D_s = \frac{a P_s}{P}$$

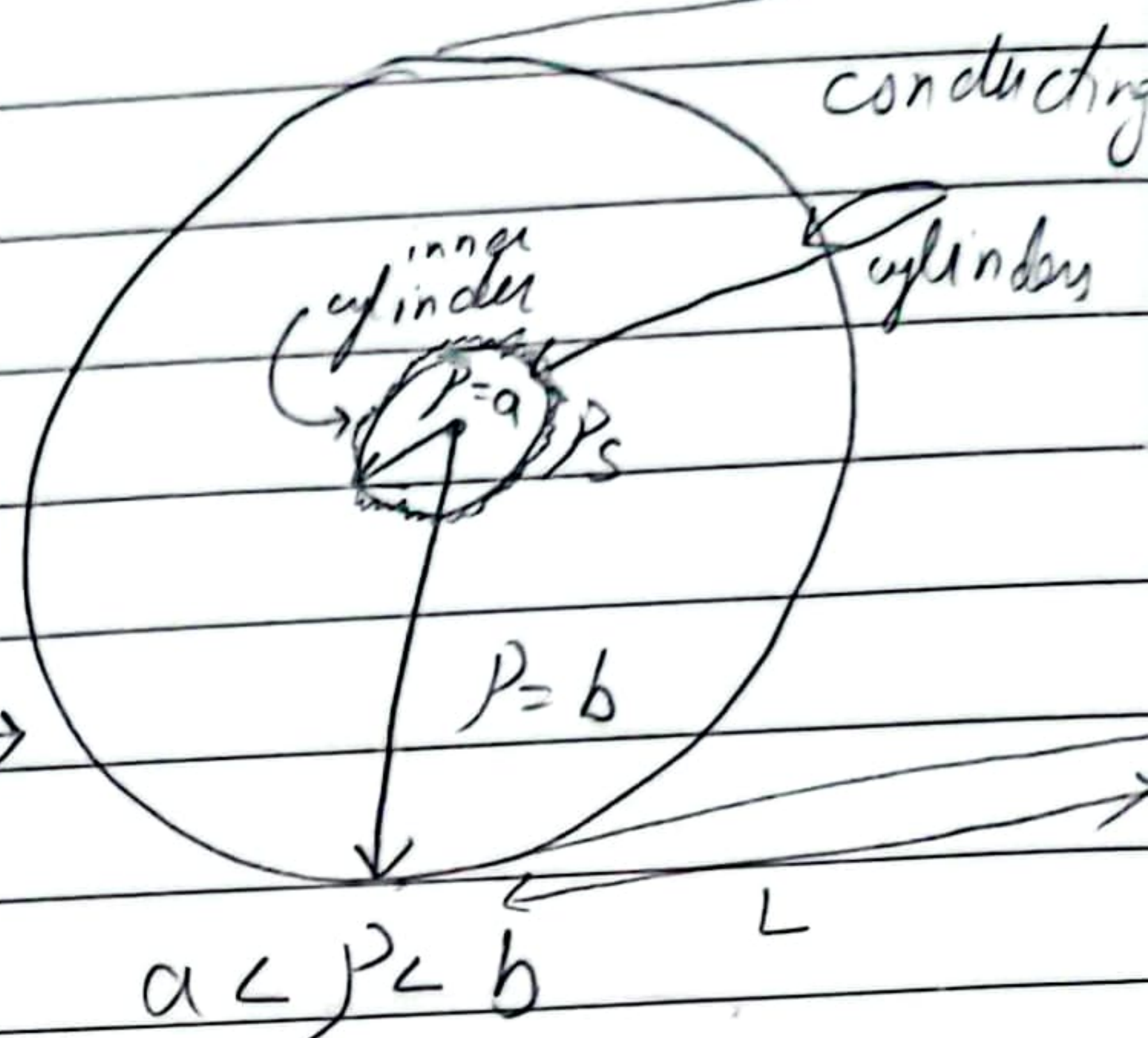
$P \rightarrow$ imaginary cylinder radius

$$Q_{\text{enclose inner}} = 2\pi a L P_s$$

$$Q_{\text{enclose outer}} = -Q_{\text{enclose inner}}$$

$$2\pi b L P_{s_{\text{outer}}} = -2\pi a L P_{s_{\text{inner}}}$$

$$\frac{P_{s_{\text{outer}}}}{P_{s_{\text{inner}}}} = \frac{-a}{b}$$



$$D_s = D_p = \frac{Q}{2\pi PL}$$

Also for line charge

$$Q = \rho_L L$$

then

$$D_s = D_p = \frac{\rho_L \cdot L}{2\pi PL} = \frac{\rho_L}{2\pi P}$$

$$D_p = \frac{\rho_L}{2\pi P}$$

$$E = \frac{\rho_L}{2\pi\epsilon_0 P}$$

$$\because D = E\epsilon_0$$

For sphere:-

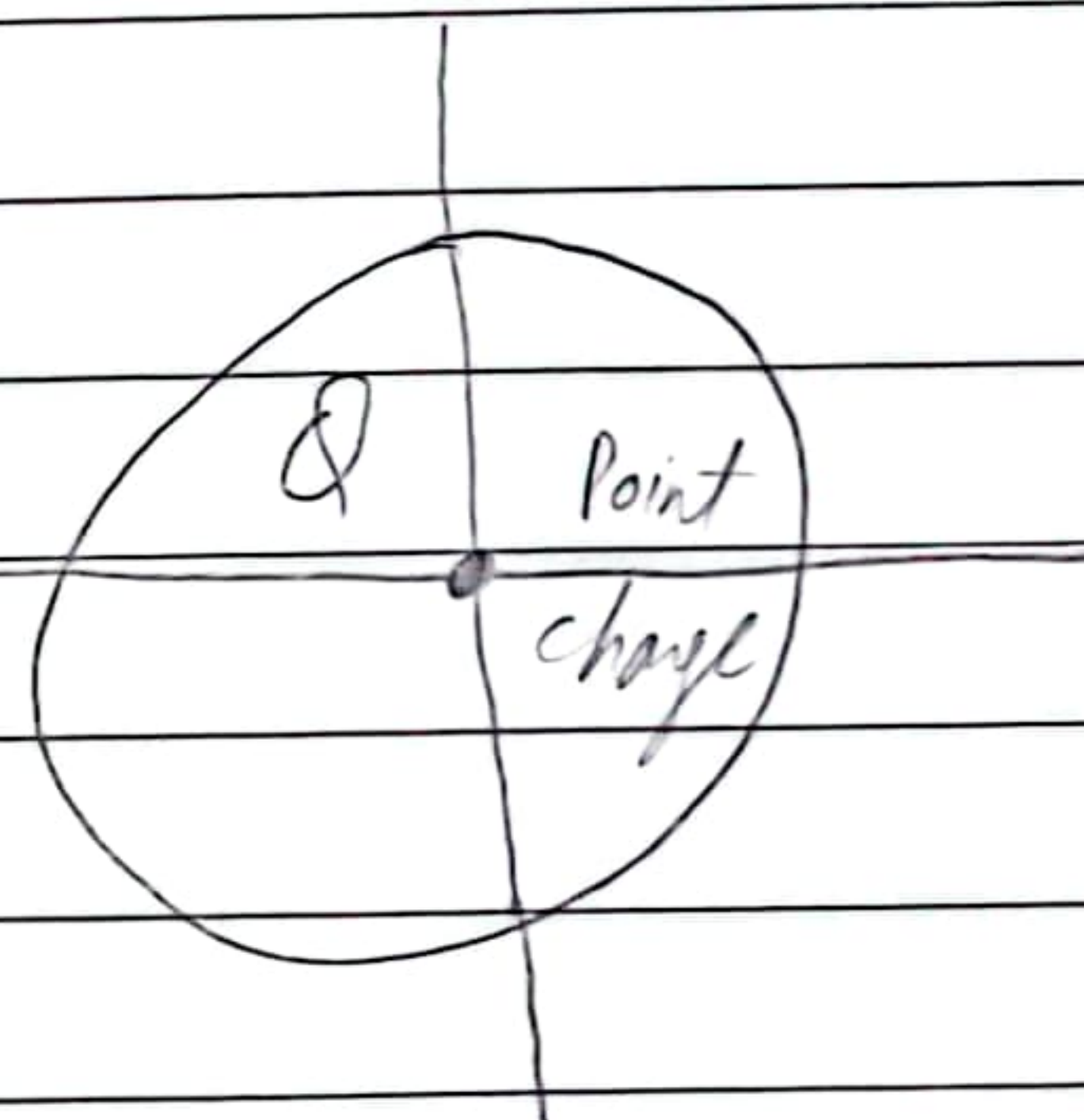
$$Q = \oint D_s ds$$

$$= D_s \oint ds$$

$$Q = D_s 4\pi r^2$$

$$D_s = \frac{Q}{4\pi r^2} \hat{a}_r$$

$$E = \frac{Q}{4\pi\epsilon_0 r^2} \hat{a}_r$$



This allows us to pull ρ outside of the integral, leaving only the surface area inside the integral.

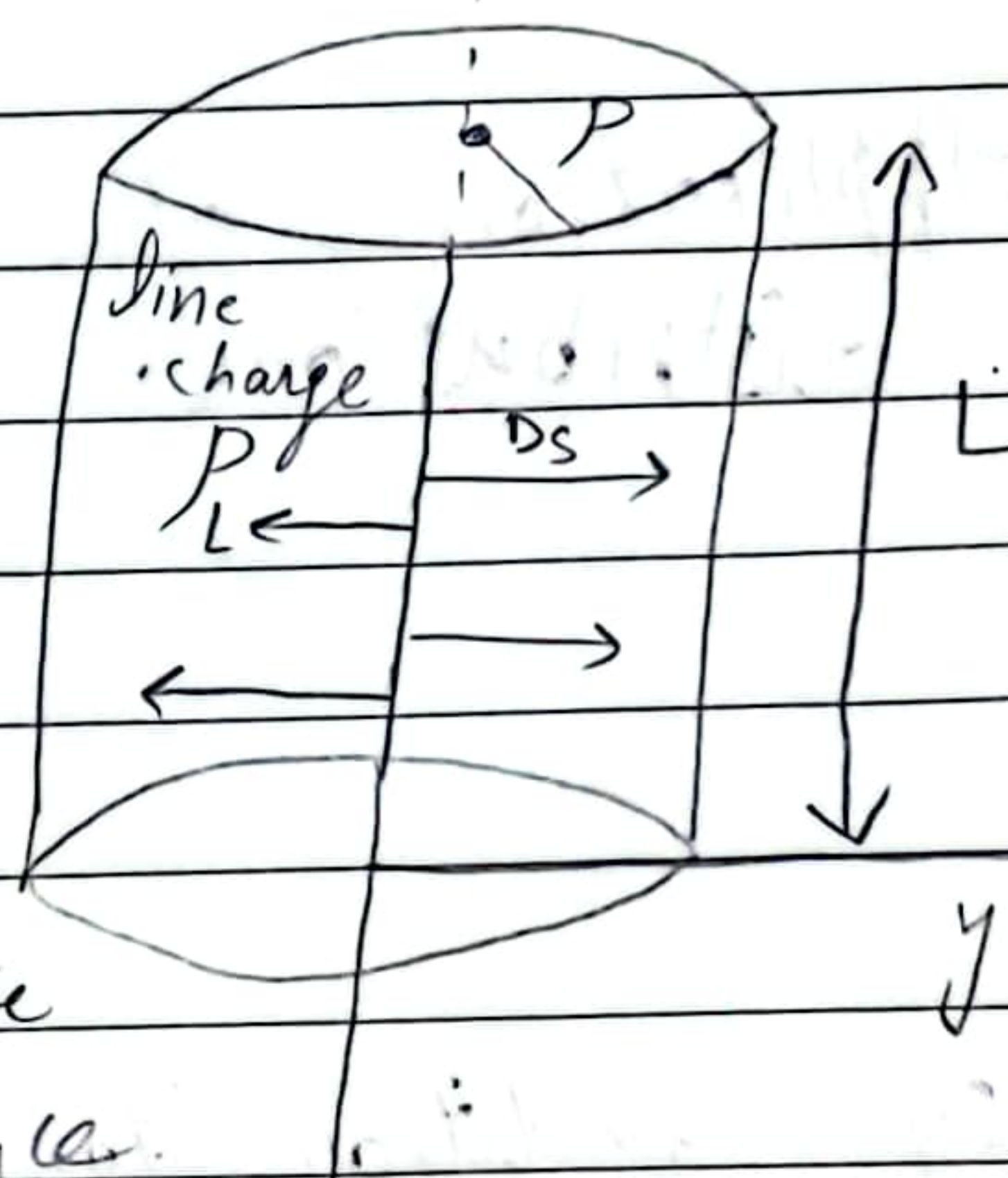
$$Q = \oint \mathbf{D_s} \cdot d\mathbf{s} = D_s \int d\mathbf{s}$$

$$= D_s \int_{\text{Top}} d\mathbf{s} + D_s \int_{\text{Bottom}} d\mathbf{s} +$$

$$D_s \int_{\text{sides}} d\mathbf{s}$$

~~=~~ no electric flux density on top and bottom cause there surfaces.

surface of an infinite line charge
 D is constant in magnitude wherever perpendicular to the cylindrical surface.



no charge is passing through these surfaces.

$$= \int_{\text{Top}} 0 d\mathbf{s} + \int_{\text{bottom}} 0 d\mathbf{s} + D_s \int_{\text{sides}} d\mathbf{s} = D_s \int_{z=0}^L \int_{\phi=0}^{2\pi} \rho d\phi dz$$

$$= D_s 2\pi \rho L$$

$$Q = D_s 2\pi \rho L =$$

$$D_s = \frac{Q}{2\pi \rho L}$$

$$b) E = \frac{Q}{4\pi\epsilon_0 r^2}$$

$$Q = \frac{Q}{4\pi r^2} \text{ or}$$

$$\frac{(0.3 \times 10^{-9}) \times 4\pi r^2}{4\pi r^2} = Q$$

$$Q = (0.3 \times 10^{-9}) (3)^2 \times 4\pi \times (3)^2$$

$$Q = 3.054 \times 10^{-7} \text{ coulombs} = 305.4 \text{ nC}$$

$$c) D = \frac{Q}{4\pi r^2} \text{ or}$$

$$E = \frac{Q}{4\pi\epsilon_0 r^2}$$

$$= \frac{3.054 \times 10^{-7}}{4\pi (4)^2} \hat{a}_r$$

$$(0.3 \times 10^{-9}) (4)^2 \times 4\pi (4)^2 = Q$$

$$D = 1.519 \hat{a}_r \text{ nC/m}^2$$

$$Q = 965 \text{ nC}$$

$$\gamma \propto Q$$

$$\gamma = 965 \text{ nC}$$